A Deep Learning-Based System for Plant Multi-Classification

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By

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<u>CERTIFICATE</u>

This is to certify that the thesis entitled **"A Deep Learning-Based System for Plant Multi-Classification"** by **"Maruf Ahmmed Munna", ID No.: 0812210205101008, Session: 2021-2022, "Md. Mustafizur Rahman Sarkar", ID No.: 0812210205101036, Session: 2021-2022, "S.A.M. Rafiul Ahsan Roly", ID No.: 0812120205101014, Session: 2021-2022,** has been accepted as satisfactory in partial fulfilment of the requirement for the degree of Bachelor of Science in Computer Science and Engineering on **Month, 2026.**

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DECLARATION

We hereby declare that our thesis entitled **"A Deep Learning-Based System for Plant Multi-Classification"** is the result of our noble work. We also ensure that it was not previously submitted or published elsewhere for the award of any degree or diploma.

The work has been accepted for the degree of Bachelor of Science in Computer Science and Engineering at Bangladesh Army University of Engineering & Technology (BAUET).

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ABSTRACT

Plants are essential for sustaining life on Earth through their production of food and their efforts in keeping our environment balanced. It is essential that we identify plants with accuracy to ensure they are properly recorded to provide accurate data related to agriculture, research, and conservation. The use of conventional techniques to identify plants relies heavily on human expertise which can be very time-consuming and tedious, as well as oftentimes inaccurate, particularly regarding similar types of plants or the early identification of plant diseases. The following research presents a discussion concerning the automatic identification of plants with the use of deep learning methods. A model for deep learning has been created to identify plants from images through the following parameters or factors plant name, organ type (roots, seeds, flowers, leaves, and fruits), plant species, edible status of the plant, and health status of the plant (healthy or diseased). The goal of the research is to provide a method by which to train and test deep learning models on a predefined set of plant images. The results indicate that deep learning models can increase the accuracy of identifying plants and provide an understanding of plant characteristics that would support future research efforts on plant monitoring, precision agriculture, and digital botany studies.

Keywords: Deep Learning, ConvNeXtBase, Plant Multi-Classification, Precision agriculture, Plant organ recognition

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Abbreviation

No.	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	DL	Deep Learning
4	CNN	Convolutional Neural Network
5	ELM	Extreme Learning Machine
6	ViT	Vision Transformer
7	YOLO	You Only Look Once
8	ResNet	Residual Neural Network
9	DenseNet	Densely Connected Convolutional Network
10	MobileNet	Mobile Neural Network Architecture
11	GNN	Graph Neural Network
12	GPU	Graphics Processing Unit
13	CPU	Central Processing Unit
14	ReLU	Rectified Linear Unit
15	GELU	Gaussian Error Linear Unit
16	ROC	Receiver Operating Characteristic
17	AUC	Area Under the Curve
18	SVM	Support Vector Machine

Chapter 1
Introduction

1.1 Introduction

Plants have an essential role in maintaining life on Earth. They provide food, oxygen, raw materials for manufacturing industries and medicines, and services that maintain or help restore the environment. Proper classification of plants is necessary for various applications, such as agriculture, biological diversity, scientific investigation, and safety measures in consumption of plants. Current techniques include morphological observation of plant structures, which requires expertise and much effort. These methods become inaccurate when dealing with species similarity, big datasets, or diseases detection at the initial stages. This leads to financial losses and jeopardizes food security.

Deep Learning (DL) can help overcome these limitations by analyzing visual patterns automatically. With proper processing of information, a DL technique can classify plant species, their organs, whether the plant is edible, and determine its health condition. In this study, a complete DL system for plant identification will be developed. The thesis will include a pipeline of activities related to obtaining, pre-processing, training, and testing the DL model.

1.2 Motivation

The complexity of traditional identification methods and the high likelihood of making errors highlight the need for automating traditional identification techniques so that the risk of misidentification, which could result in decreased agricultural production, damage to farmers' finances and degradation of biodiversity. Furthermore, misclassification can negatively impact the success of future conservation efforts. To this end, DL technology presents a tremendous opportunity to accurately identify various types of entities within an image and develop systems to classify, confirm, or reject the identity of any organisms or items present in an image. Consequently, this study assesses the incorporation of DL technology into machine-learning models that can provide automated identification of organisms or items in the agricultural and environmental monitoring industries.

1.3 Rationale of the Study

The identification of plants accurately is key to maximizing their agricultural productivity, conducting accurate biodiversity research, and protecting habitats to achieve their goals for farmers, scientists, and conservationists. Traditional techniques have limitations with regards to visually similar species, as well as requiring an expert to perform the identification. Advancements in the areas of digital imaging and DL allow for computer-based recognition of visual characteristics of plant specimens throughout thousands of images. Although some commercial systems have been developed, none completely incorporates vegetative structure classification, species classification, whether plants are edible, and the overall health of plants. The purpose of this project is to bridge the gap in the field of plant identification by developing a comprehensive DL based system that improves accuracy, reduces manual input, and supports the future of agriculture and botany in a modern environment.

1.4 Problem Statement

Accurate identification of plants and their respective characteristics is critical in agriculture, conservation, and botanical society research. Conventional methods of plant identification depend on an expert's evaluation of plant morphology (leaves, flowers, seeds, roots, fruits), which requires a high level of time, skill, and experience. Misidentification often happens due to similar appearances in the same family, which leads to incorrect classifications and therefore poor decision making.

For example, if errors occur in plant identification for agriculture, the plant species or diseases could not be identified earlier; this may reduce yield potential and hence affect food security for the country. Similarly, if errors occur in research and conservation, it would be difficult to accurately identify and manage resources and to assess biodiversity.

Current methods of plant identification are a combination of conventional and deep learned systems, yet most have focused on a single use case, such as identifying species, disease, and/or an individual's characteristics like organ types, edible, etc. For this reason, this research aims to develop an automated deep learning model to correct these limitations. It will analyze images of plants and classify the organ types (leaves, flowers, seeds, roots) along with the species and edibility, and overall health status of each plant, providing for a complete and comprehensive accurate identification.

1.5 Objectives

This research aims to develop a system that uses deep learning for automatic plant identification from images by defining the research objectives, as well as providing the means to assess them.

Main Study Objectives

The overarching goal of this research is to evaluate whether image-based plant recognition and classification using deep learning is effective, accurate, and beneficial, relative to how plants are typically annotated, classified, and identified. Specifically, the study aims to find out whether plant identification and classification using deep learning can be done successfully and efficiently compared with traditional way of identifying plants.

Specific Study Objectives

- **To create a comprehensive DL model:** Utilizing a single model to classify all species and organs, and to enable recognition of new, unseen photos based on the features learned from the training images.

- **To identify the organ type and species:** Enabling the model to identify organ types and species by using visual characteristics to distinguish between organ types and specific species.

- **To identify diseased or healthy plants:** Enabling the model to differentiate between diseased and healthy plants by recognizing differences that indicate a diseased plant.

- **To build a structured dataset:** Creating a large collection of images, with a variety of species represented, which have been pre-processed and are divided by species, organ type, edibility, and health status.

- **To measure the model's performance:** To determine the effectiveness of the DL model for each plant by calculating its performance using the following metrics: 1) accuracy, 2) precision, 3) recall, 4) F-1 score, 5) confusion matrix, 6) classification report; 7) accuracy plots; and 8) ROC curves; to verify reliable classification or identification of the plants.

1.6 Addressing Course Outcomes (CO), Program Outcomes (PO) and Complex Engineering Problems & Activities

Table 1.6.1: CO, PO, Knowledge Profiles (K), Complex Engineering Problems (P) and Complex Engineering Activities (A) Mapping

COs	**Mapped Program Outcomes (PO)**	**Mapped Knowledge Profiles (KP)**	**Complex Engineering Problems (P)**	**Complex Engineering Activities (A)**
CO1	PO1, PO2, PO4	K1, K2, K3, K8	P1, P2	A1
CO2	PO2, PO4	K4, K8	P1, P3	A2
CO3	PO3	K5	P2, P4, P5, P7	A1, A3
CO4	PO5	K2, K6	P2, P6	A3
CO5	PO4, PO5	K2, K6, K8	P1, P3, P7	A1, A3
CO6	PO8, PO10	K9	P5, P6	A1, A2
CO7	PO9	K7, K9	P6	A4
CO8	PO6, PO7	K5, K7, K9	P2, P5, P6	A4
CO9	PO11, PO12	K8	P1	A3, A5

Table 1.6.2: Mapping of COs with BAETE Program Outcomes (POs)

COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1												
CO2												
CO3												
CO4												
CO5												
CO6												
CO7												
CO8												
CO9												

Table 1.6.3: Mapping of COs with Knowledge Profiles (K1-K9)

COs	K1	K2	K3	K4	K5	K6	K7	K8	K9
CO1									
CO2									
CO3									
CO4									
CO5									
CO6									
CO7									
CO8									
CO9									

Table 1.6.4: Mapping of COs with Complex Engineering Problems (P1-P7)

COs	P1	P2	P3	P4	P5	P6	P7
CO1							
CO2							
CO3							
CO4							
CO5							
CO6							
CO7							
CO8							
CO9							

Table 1.6.5: Mapping of COs with Complex Engineering Activities (A1-A5)

COs	A1	A2	A3	A4	A5
CO1					
CO2					
CO3					
CO4					

CO5					
CO6					
CO7					
CO8					
CO9					

CO1: Problem Identification & Analysis

Program Outcomes

- **PO1: Engineering Knowledge Application:** Applied knowledge of computer science fundamentals, deep learning principles, and image processing techniques to analyze plant images and design a framework for automated plant classification.
- **PO2: Problem Analysis:** Identified and analyzed challenges in plant classification, including variability in images, similarity of species, and overlapping disease symptoms. Formulated research strategies considering dataset quality, preprocessing requirements, and classification accuracy.
- **PO4: Investigation:** Conducted systematic investigation using curated plant image datasets, including benchmarking classification accuracy, evaluating preprocessing techniques, and analyzing model performance under different conditions.

Knowledge Profiles

- **K1: Natural Sciences:** Understanding plant morphology, organ types, growth stages, and disease pathology guided dataset annotation and feature recognition.
- **K2: Mathematics:** Used statistical and probabilistic methods to evaluate model performance and optimize preprocessing algorithms.
- **K3: Engineering Fundamentals:** Applied software engineering principles for model training pipelines, dataset management, and experiment reproducibility.
- **K8: Engagement in Research:** Integrated recent studies on plant phenotyping, disease detection, and medicinal plant analysis to inform methodology.

Complex Engineering Problems

- **P1: Depth of Knowledge Required:** Combined knowledge of deep learning, image processing, and plant biology to design and implement the classification framework.

- **P2: Range of Conflicting Requirements:** Balanced high accuracy vs. computational efficiency, preprocessing complexity vs. dataset size, and model generalization vs. organ-specific recognition.

Complex Engineering Activities

- **A1: Range of Resources:** Managed plant image datasets, preprocessing pipelines, and DL model resources for efficient experimentation and evaluation.

CO2: Literature Review & Research Gap Analysis

Program Outcomes

- **PO2: Problem Analysis:** Reviewed existing plant classification and disease detection research to identify gaps in multi-organ recognition, health evaluation, and medicinal plant analysis. Defined research objectives to address these gaps.
- **PO4: Investigation:** Conducted literature review, comparative study of DL methods, and evaluation of prior datasets and models to identify limitations and research gaps.

Knowledge Profiles

- **K4: Engineering Specialization:** Applied deep learning and image processing specialization to design feature extraction and classification experiments.
- **K8: Engagement in Research:** Integrated insights from recent studies in plant phenotyping and disease analysis.

Complex Engineering Problems

- **P1: Depth of Knowledge Required:** Applied DL, image processing, and plant biology knowledge to address gaps in prior studies.
- **P3: Depth of Analysis Required:** Tackled challenges of visually similar species and overlapping disease symptoms.

Complex Engineering Activities

- **A2: Level of Interactions:** Resolved trade-offs in dataset size, image quality, preprocessing techniques, and model complexity to improve classification accuracy.

CO3: Methodology Design & Implementation

Program Outcomes

- **PO3: Design/Development of Solutions:** Designed a deep learning–based plant classification system considering multiple features such as organ type, species, edibility, medicinal uses, and health status. Ensured robust methodology for dataset preparation, preprocessing, and model evaluation.

Knowledge Profiles

- **K5: Engineering Design and Support:** Focused on scalable, modular, and maintainable model training pipelines and preprocessing workflows.

Complex Engineering Problems

- **P2: Range of Conflicting Requirements:** Balanced classification accuracy vs. computational efficiency and preprocessing complexity vs. model generalization.
- **P4: Familiarity with Issues in Practice:** Addressed challenges with variable image conditions, such as lighting and background variability.
- **P5: Extent of Applicable Codes:** Applied contemporary DL frameworks and IP techniques for plant classification.
- **P7: Interdependence:** Managed dependencies between dataset quality, model performance, and preprocessing efficiency.

Complex Engineering Activities

- **A1: Range of Resources:** Managed plant image datasets, preprocessing algorithms, and DL model resources.
- **A3: Innovation:** Introduced multi-organ recognition (roots, seeds, leaves, flowers, fruits) and integrated health and medicinal feature classification.

CO4: Modern Engineering Tools Application

Program Outcomes

- **PO5: Modern Tool Usage:** Utilized deep learning frameworks (TensorFlow/Keras) and image processing libraries (OpenCV) for preprocessing, training, and evaluation of plant images.

Knowledge Profiles

- **K2: Mathematics:** Applied statistical and mathematical modeling for performance analysis.
- **K6: Engineering Practice:** Followed best practices for reproducible experimentation, dataset management, and model evaluation.

Complex Engineering Problems

- **P2: Range of Conflicting Requirements:** Balanced accuracy, preprocessing time, and computational efficiency.
- **P6: Stakeholder Involvement and Conflicting Requirements:** Considered research needs, reproducibility, and practical usability.

Complex Engineering Activities

- **A2: Level of Interactions:** Optimized preprocessing pipelines and DL model architecture for improved classification performance.

CO5: Experimental Analysis & Data Evaluation

Program Outcomes

- **PO4: Investigation:** Conducted systematic experiments using curated plant datasets. Benchmarked model performance on multi-organ recognition, species classification, edibility, medicinal, and health features.

Knowledge Profiles

- **K2: Mathematics:** Used statistical metrics (accuracy, precision, recall, F1-score) for performance evaluation.
- **K6: Engineering Practice:** Followed rigorous experiment protocols for reproducibility and reliability.
- **K8: Engagement in Research:** Compared results with recent studies to validate methodology.

Complex Engineering Problems

- **P1: Depth of Knowledge Required:** Applied deep understanding of DL, IP, and plant biology.
- **P3: Depth of Analysis Required:** Addressed visually similar species, organ variations, and overlapping disease symptoms.
- **P7: Interdependence:** Managed dataset quality, preprocessing pipelines, and model performance interactions.

Complex Engineering Activities

- **A1: Range of Resources:** Handled datasets, preprocessing pipelines, and model training resources.
- **A3: Innovation**: Integrated multi-feature plant classification including organ type, health, and medicinal value.

CO6: Research Management & Leadership

Program Outcomes

- **PO8: Ethics:** Ensured responsible dataset usage and transparency in methodology.
- **PO10: Communication:** Maintained detailed documentation of methodology, dataset, and experimental results.

Knowledge Profiles

- **K9: Ethics and Conduct:** Ensured responsible use of datasets, fairness in model evaluation, and reproducibility of results.

Complex Engineering Problems

- **P5: Extent of Applicable Codes:** Applied DL frameworks and IP standards for reliable implementation.
- **P6: Stakeholder Involvement:** Considered researchers' needs and academic standards for reproducibility.

Complex Engineering Activities

- **A1: Range of Resources:** Managed datasets, DL pipelines, and experimental resources.
- **A2: Level of Interactions:** Coordinated preprocessing, model training, and evaluation activities.

CO7: Effective Communication & Presentation

Program Outcomes

- **PO9: Individual & Team Work:** Prepared reports, visualizations, and methodology diagrams to communicate research findings effectively.

Knowledge Profiles

- **K7: Role of Engineering:** Demonstrated societal impact through accurate plant classification and disease detection.
- **K9: Ethics and Conduct:** Ensured clarity, transparency, and accessibility in reporting results.

Complex Engineering Problems

- **P6: Stakeholder Involvement:** Addressed needs of researchers and academicians.

Complex Engineering Activities

- **A4: Consequences to Society:** Contributed to sustainable agriculture and improved research practices.

CO8: Ethical & Social Impact Recognition

Program Outcomes

- **PO6: Engineer & Society:** Evaluated societal implications of plant classification research, emphasizing accessibility, fairness, and academic responsibility.
- **PO7: Environment & Sustainability:** Promoted sustainable computing practices by optimizing dataset usage and preprocessing pipelines.

Knowledge Profiles

- **K5: Engineering Design and Support:** Designed reproducible, scalable, and maintainable DL models and pipelines.
- **K7: Role of Engineering:** Supported sustainable agricultural practices and research reliability.
- **K9: Ethics and Conduct:** Ensured fairness and ethical handling of datasets.

Complex Engineering Problems

- **P2: Range of Conflicting Requirements:** Balanced accuracy, computational cost, and usability.
- **P5 & P6: Extent of Applicable Codes and Stakeholder Involvement:** Applied DL standards and considered multiple academic and research requirements.

Complex Engineering Activities

- **A4: Consequences to Society:** Contributed to research reproducibility, sustainable agriculture, and accurate plant identification.

CO9: Lifelong Learning & Research Adaptation

Program Outcomes

- **PO11: Project Management:** Applied structured research methodology, iterative experimentation, and systematic dataset handling.
- **PO12: Lifelong Learning:** Demonstrated continuous learning in DL, image processing, and plant classification research methods.

Knowledge Profiles

- **K8: Engagement in Research:** Integrated recent findings in plant phenotyping, disease detection, and multi-organ classification.

Complex Engineering Problems

- **P1: Depth of Knowledge Required:** Combined DL, IP, and plant biology knowledge for research innovation.

Complex Engineering Activities

- **A3: Innovation:** Developed multi-organ, multi-feature plant classification methodology.
- **A5: Familiarity:** Gained expertise in dataset handling, DL model development, and plant image analysis.

1.7 Report Layout

The report is structured into five distinct chapters that collectively address different facets of the research into DL based plant identification and classification. The overall aim of the report is to present the research problem systematically and explain the methodology to be used in analyzing experimental results, as well as to draw conclusions regarding findings from the research and directions for future research.

Chapter 1: Introduction: This chapter introduces the problem of plant identification and explains the need for automatic ways of identifying plants in agriculture and botanical research, and for identifying plants for environmental monitoring. It provides justification for the research being conducted, identifies the objectives of the research, defines the scope of the research and outlines the contributions to be made by the system described in this report, and discusses how the research aligns with Course Outcome (CO), Program Outcome (PO) and Complex Engineering Problems and Activities.

Chapter 2: Background Study: This chapter is a literature review of existing knowledge related to plant identification and plant disease detection as well as image-based plant classification. It examines traditional methods of identifying plants and examines deep learning techniques that have been utilized for analyzing images of plants and describes the limitations that exist in these areas of study to develop research gaps that support the need for conducting this research.

Chapter 3: Methodology: This section discusses the approach taken to create a deep-learning system for identifying plants through the collection of data, pre-processing the data, preparing images, and building the model. In addition to these activities, this chapter will describe the specific architecture of the deep-learning model that predicts which species the plant is, how to identify specific parts of the plant, whether the plant is edible, and if the plant is in good or bad condition. Also described will be the methods used to train the models, set hyperparameters, and implement them in practice.

Chapter 4: Results and Discussion: In this chapter, the results of the experiments performed will be presented. The performance of the new system will be assessed using performance metrics such as accuracy, precision, recall and F1-score against data obtained through various assessments. In addition, the results obtained from these metrics will be analyzed and discussed to provide a basis for establishing the effectiveness of the deep learning model in identifying and classifying plants.

Chapter 5: Conclusion and Future Work: This chapter will summarize the overall findings and contributions of this thesis and will reiterate how effective the proposed deep learning-based method is at automatically identifying and classifying plants. Additionally, this chapter will discuss possible future directions for this work, including future improvements of the system's accuracy, the expansion of the dataset by adding additional types of plants, and creating real-time applications for monitoring agricultural production and smart farming systems.

1.8 Conclusion

This chapter provided an overview of the research associated with deep learning methods in relation to identifying and classifying plants. It also discussed the significance of being able to properly identify plants for purposes such as agriculture, preservation of the environment and botany. The chapter outlined the limitations associated with traditional manual identification methods and indicated the need for automatic systems that can efficiently perform analyses of plant images. To substantiate the significance of creating an automatic plant identification system, a detailed description of the motivation, rationale for the study, and the problem statement of the study were given. The research objectives were also identified to help guide the creation of a deep learning framework that could identify plant species, identify the types of organs for those species, determine the species’ edibility and locate indications of poor health in plant species.

Chapter 2
Background Study

2.1 Introduction

Multi-organ plant classification systems based on CNN have attained accurate rates over 92%, enabling effective arrangement of leaves, flowers, roots, seeds and fruits [3].

Besides CNNs, vision transformer (ViT) architectures capture the detailed global contextual information throughout the image which can detect subtle differences between species or early symptoms of diseases [6].

In applications spanning agriculture, environmental conservation, botanical exploration and food security [10], knowing what plant species are present in each environment, as well as their characteristics are of vital importance. Examples include separating between two closely related Solanaceae species based on leaf morphology, which is often not reliable and has repercussions both in crop management as well as research accuracy. This manual identification process also fails to scale or account for diversity. The world has more than 390,000 known species of plants, and the network for recording biodiversity is vast. Human inspection is typically time-consuming such that, in the agricultural sector, crop yield and economic returns can be lowered by delayed detection of diseases or stress conditions because human cannot detect plants' abnormal condition timely [10]. Such limitations underscore the need for automated, scalable solutions that can process large datasets of plant images efficiently and with accuracy across species, organs and health conditions.

Plants are the building blocks of life on Earth, offering up the essentials for all living things to flourish. They provide food and oxygen, participate in the carbon cycle, and function as primary producers within ecosystems. Besides their ecological function, plants provide medicinal compounds, raw materials for construction and industry, pharmaceuticals and biotechnology resources [15]. The simultaneous prediction of species, organ type, health status and edibility in multi-task learning frameworks greatly enhances system capabilities, thus providing a more holistic method for automated plant care [15].

DL has led to rapid transformations in plant identification. Among deep learning models can automatically access and learn low level textures, mid-level organ pattern and high-level species-specific features from images in a hierarchical order [16]. These models minimize the need for hand-crafted features allowing precise classification of plant organs and species as well as determination of disease state and edibility. These systems facilitate precision agriculture, ensuring timely disease identification, efficient crop processing and effective use of resources. They also enable digital botany, allowing mass biodiversity surveys and automation of conservation. In addition, robotic plant recognition systems help both sustainable farming and environmental conservation. Accurate monitoring of species and health allows for timely intervention, helps to prevent overuse of pesticides, and reduces losses to crops. These systems also enhance food safety and quality by effectively distinguishing between edible and toxic plants. Such automated plant monitoring and identification has become essential to maintaining ecological equilibrium, agricultural proficiency, and human health as the stresses of climate change, population growth, and food demand continue to increase [16].

Traditionally, identification was done through the manual observation of morphological features such as leaf shape, flower structure, seed and fruit characteristics as well as root architecture. Identifying species accurately required extensive expertise on the part of botanists and agricultural specialists. And although these methods have been the basis for botanic sciences, they are tedious processes that require significant investment in manpower and time, making it easy to mistake visually similar species or plants showing symptoms of novel pathogen diseases at its early stages [28].

2.2 Literature Review

A literature review is a critical synthesis of existing research on a specific topic, acting as the foundation for a thesis or research paper. Rather than a simple summary of past studies, it analyzes, compares, and evaluates published literature to highlight major trends, debates, and themes. By mapping out the current state of knowledge, it identifies gaps or contradictions in previous work that your own study aims to address. Ultimately, this section contextualizes your research, proving its relevance and justifying your methodology to the academic community.

2.2.1 Traditional Plant Identification Methods

The extracted features were used to classify with classical machine learning algorithms, including support vector machines (SVMs), k-nearest neighbors (k-NN), and random forests. Although they did improve classification performance, reported accuracy ranged from 64% to

92%, dependent on dataset size, organ type and feature selection [1]. These methods can only detect a small number of species or organs at once, and are poorly suited for multi-organ, multi-species or disease detection.

Even trained botanists could have common difficulty identifying species with very similar morphology, particularly within genera where leaf shape or flower form varied only slightly [10]. The challenges of manual identification became increasingly a point of contention, at the time, due to increased motivation need to catalog and identify hundreds of thousands of plant specimens per year for agriculture, conservation, research. Early detection of diseases and health monitoring were indeed a challenge, since subtle lesions, color changes or deformities in an organ could easily be missed. However, without the presence of flowers either differentiation seen in two species belonging to the same family can lead to its misidentification, these specific adaptations should be used along with neighboring morphological features for proper disambiguation [10].

The traditional methods of identifying plants were the basis for taxonomy and classification, but they had major shortcomings in terms of scalability, consistency, and speed. The dependence on highly skilled expertise, subjective experience, and morphological variability opened the way for automated and data-driven methodologies. These restrictions acted as the direct propeller behind utilizing deep learning approaches to be able to learn hierarchical visual features directly from images of plants and their organs, allowing for precise classification of multi-organ from different varieties by various diseases [16].

While these systems lessened the manual workload and made quantitative comparisons possible, their performance was limited. Factors like variability in lighting, background clutter, partial occlusion, and complex organ shapes frequently resulted in misclassification. And these approaches relied heavily on hand-crafted features that failed to fully represent the diversity of plant phenotype [26].

The process of identifying plant species has a long tradition dating back to the early era of botanical studies, when species were primarily identified through potentially observable morphological characteristics. Speciation traditionally depended on detailed observation of morphology, including leaves, flowers, fruits and seeds, roots [28]. Leaf morphology (shape, size, margin, venation, arrangement and texture) was one of the most widely adopted traits owing to its ease of obtaining and occurrence throughout the growing season. One of the most important characteristics, though, was flower structure: petal number, symmetry and coloration as well as reproductive structures were vital for species identification. Fruits and seeds were analyzed in terms of shape, size, texture, and dehiscence patterns; roots and stems were examined for structure, branching patterns; internal anatomy [28]. As digital image-processing technologies developed, plant identification was attempted to be made automatic by researchers. The approaches used were color thresholding, edge detection, shape descriptors, and texture analysis on digitized leaves, flowers and fruits. Computer-assisted systems have measured features such as leaf area, perimeter, circularity, vein density and color histograms for species classification [28].

2.2.2 Deep Learning for Plant Classification

DL based plant classification systems predominantly rely on Convolutional Neural Networks (CNNs). A CNN has several component layers: convolutional layers to extract local spatial features with learned filters, activation functions such as ReLU to add non-linearity, pooling layers to reduce dimensionality and ensure spatial invariance, and fully connected layer for the classification task based on extracted feature [3]. CNNs automatically identify edges, textures, and organ shapes as well as disease patterns from images of leaves, flowers, fruits, seeds, and roots. CNN's multi-layer feature extraction ability can distinguish visually similar species that features engineering struggled with.

A more recent approach to classifying plant species with deep learning is Vision Transformers (ViTs). In contrast to CNNs with local convolutions, ViTs divide an image into fixed-size patches and encode them in linear embeddings. Subsequently, self-attention layers model global contextual relationships between the patches, enabling the model to learn long range dependencies as well as subtle variations across organs, species or disease manifestations [6]. Interestingly, ViTs have been shown to be robust backbone networks for intricate classification tasks, including differentiation between plant species with almost identical foliage or early disease recognition on small areas of organs. The optimized ViT models have already reached an accuracy of 94% for benchmark multi-organ datasets [6]. In ViT base models, it is observed that they performed better in fine-grained classification issues like early disease detection or differentiating subtle variations in leaves morphology [6].

Deep learning systems have been expanded to multi-task learning frameworks that allow prediction of multiple attributes simultaneously: species, organ type, health status, and edibility [15]. Here, the shared convolutional or transformer layers learn general features, and each task-specific head predicts for each attribute. Introducing multi-task learning also reduces both training time and prediction variance, since knowledge should easily transfer from one task (such as organ type) to another (such as correct species recognition).

Traditional plant identification and early computer-aided methods had limitations (reliance on handcrafted features and sensitivity to variation in the environment), which motivated the use of DL techniques for automated plant classification [16]. DL models learn hierarchical feature representations from raw images that are complex, including multi-organ, multi-species and various disease types and are not feature engineered. Multi-task DL models have an advantage over single-task models, especially in complex multi-class datasets with hundreds of images from various species and organ types [16]. That's the gist of it; deep learning is a powerful, scalable and accurate approach to classifying plant species. Hierarchical feature extraction is enabled via CNNs and their variants, transfer learning uses available knowledge for efficient training, ViTs capture global contextual information well, while multi-task learning performs concurrent prediction of multiple plant attributes. These features overcome the limitations of conventional approaches and make the development of effective systems for detecting plants in agriculture and research domains possible [16].

Deep learning models achieved in practice with high accuracy. Multi-organ plant classification systems based on CNNs exceeded 92% accuracy and disease detection models utilizing attention-augmented CNNs obtained over 91% F1-scores [17].

An effective way in dealing with this data scarcity, especially for plant datasets, has become transfer learning. Networks like ResNet, DenseNet and EfficientNet are pre-trained on datasets (like ImageNet), therefore they learn generalized feature representations [29]. More recently, training these models on plant-specific datasets facilitates this quick adaptation, and increases classification performance from a few training images. For instance, EfficientNetB3 provided an accuracy of more than 93% for multi-organ and multi-species classification tasks involving thousands of images in datasets across more than 20 different species of plants [29]. Additionally, transfer learning helps to lower computational load and prevents overfitting from occurring in networks trained on rare organs or cases of diseases. As well as achieving better classification, DL models are robust in the face of differences in lighting, occlusion and background complexity which often affect traditional methods. These data augmentation techniques rotation, scaling, flipping and color jittering can also increase the model generalization by imitating real-world field conditions [29]. High-throughput imaging of leaves, flowers, fruits and seeds for species identification, health assessment and edibility are possible in real time when a multi-organ DL model is encapsulated along with drones or mobile devices. This ability is crucial for precision agriculture, early disease intervention, biodiversity studies and conservation programs.

2.2.3 Disease Detection and Plant Phenotyping

DL based plant classification systems predominantly rely on Convolutional Neural Networks (CNNs). A CNN has several component layers: convolutional layers to extract local spatial features with learned filters, activation functions such as ReLU to add non-linearity, pooling layers to reduce dimensionality and ensure spatial invariance, and fully connected layer for the classification task based on extracted feature [3]. CNNs automatically identify edges, textures, and organ shapes as well as disease patterns from images of leaves, flowers, fruits, seeds, and roots. CNN's multi-layer feature extraction ability can distinguish visually similar species that features engineering struggled with.

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2.3 Comparative Study

The comparison section of the study covers task formulation, model architecture, dataset characteristics and properties, interpretability with respect to explainability including visualization methods as well as potential issues when deploying in practice asking for plant classification, organ identification and disease detection along without access to multi-task learning. It highlights trends, performance and research gaps based on the 30 papers.

2.3.1 Task Formulation: Classification vs Detection vs Multi-task Learning

Multi-task learning combines classification and detection tasks, enabling predictions of organ type, species, health status, and edibility based on a single image [15]. Multi-task learning helps prevent wasteful computation, as well as exploit shared features to improve generalization and provide consistency among predictions. It is known from studies that multi-task architecture outperforms single task models, especially for hundreds of images with multiple organ types [15].

Detection combines the spatial dimension, determining where organs or diseased areas are discovered within images. Region-based CNN or YOLO frameworks can find multiple organs at the same time as well as healthy and diseased areas [16]. Detection is paramount for phenotyping and early disease management as it facilitates organ-specific intervention.

The most prevalent method of organ identification is classification, which involves labeling plant species or organs. EfficientNetB3 trained on multi-organ datasets, for example, obtained species recognition accuracy of 92-94% [29]. Classification is efficient in computation and can be used for applications where exact localization of organs is not required.

2.3.2 Model Families and Architectural Trends

CNN backbones have been extensively used for plant classification and disease detection because they can learn hierarchical visual features. For example, ResNet employs skip connections to enable deeper networks without vanishing gradients and DenseNet combines feature maps from all previous layers, which promotes feature reuse [3].

Vision Transformers (ViTs) use self-attention to capture global dependencies and subtle differences across the plant organs. ViTs outperform local CNN filters for multi-organ and disease detection tasks when differences are less visually apparent [6].

MobileNet or EfficientNet provide a practical trade-off suitable for real-time deployment and automated monitoring using edge devices or drones. Although they are less accurate than transformer-based models, they enable scalable and energy-efficient applications in the area of AI [7].

Hoya-CNN2GNN is a hybrid CNN-GNN architecture in which the convolutional features are extracted and combined with a graph representation of relationships existing between plant organs [13]. This is particularly valuable for multi-organ phenotyping as it allows spatial context between leaves, flowers, and fruits to enhance classification.

2.3.3 Dataset Limitations and Generalization Issues

Cross-species generalization is another limitation. A model trained on a subset of species is unlikely to generalize to otherwise unseen species without transfer learning or domain adaptation. Leaf shape, flower architecture or organ color vary across species and can be misclassified if a model has not seen the same patterns during training [1]. Data annotation is another bottleneck. It takes an expert to accurately label organs, species, diseases and edibility. For instance, separating closely related Solanaceae species based on leaf venation or flower morphology usually requires input from a botanist. There is limited diversity in dataset and model performance due to expert annotations [1].

Moreover, such variants to obtain images in different environments and training on field-acclamation can mitigate the domain shift and enhance generalization [7].

Class imbalance is a big challenge. In multi-organ datasets, leaves are often overrepresented and flowers, fruits, and roots underrepresented. Common diseases occur frequently in datasets and rare or early-stage diseases are few. Such an imbalance would lead the model to overfit abundant classes and hence low recall for sparse classes, which is mostly undesirable in disease diagnostics where early-stage identification is of utmost importance [14].

Data scarcity, however, may be circumvented through multi-task learning, where models exploit commonalities in representations between different organs or species [15].

Deep learning models performance towards plant classification, organ identification and disease detection is greatly reliant on dataset quality, size and diversity. Although several plant-related image datasets are publicly available, many of those contain a very small number of species and plant organs or disease types. As an example, while the PlantCLEF datasets include thousands of images, they are typically dominated by common species, and there is little representation of rare or endangered species [20].

Domain shift reflects the deteriorating performance of a model on images captured under conditions different from those of its training data. Lab-trained or controlled images projected and exploited models, usually yielding failed predictions in the actual field as [28] have revealed extreme variations of features such as differences on lighting, backgrounds, occlusion scenes-of-motion or weather vagaries. For instance, a CNN trained to predict leaf diseases with clear or unobstructed images may confuse the same leaf image as diseased in case it was captured under a partial shade or dappled sunlight.

Overcoming these limitations is essential for practical deployment. These strategies can be data augmentation (rotation, flipping, scaling, color jittering), transfer learning, domain adaptation and synthetic data generation to increase enrichment of rare classes [29].

The primary studies discussed in this chapter are summarized in Table 1. The methods employed, the characteristics of the data sets used, the tasks undertaken, the number of classes created, and the reported accuracy are all shown together in a single table. These consolidated data allow for a convenient examination of the trends in performance obtained by different methods, either through deep learning or hybrid techniques.

Table 2.3.1: Summary of Methods, Datasets, and Accuracy for Reviewed Plant Classification Studies

Methods Used	**Dataset Type**	**Task**	**Classes & Accuracy**
Review of CNN, transfer learning,	Review of public and custom	Review of medicinal plant identification	N/A

hybrid models, transformers **[1]**	medicinal-plant image datasets		
Lightweight custom CNN + OpenCV/NumPy **[2]**	Public PlantVillage leaf image dataset	Automated plant disease detection	2 classes, 92.06%
Attention-Augmented Residual Network, CGAN **[3]**	Public PlantVillage dataset	Plant disease classification/detection	38 classes, 98.78%
YOLO11 + SAM + ResNet-50 + LIME **[4]**	PlantVillage + PlantDoc tomato datasets	Tomato leaf disease identification	9 classes, 55.33%
VIGAM (Real-ESRGAN / EDSR) + ResNet/ViT + GPT-2 **[5]**	PlantVillage-derived and drone-style image setting	Drone-based plant disease diagnosis	38 classes, Best: Real-ESRGAN + ViT
Optimized Vision Transformers **[6]**	PlantVillage dataset	Plant disease detection/classification	38 classes, 99.77%
Transfer learning with MobileNet-V3 (multi-organ) **[7]**	Custom mangrove dataset (leaves, stems, seeds)	Mangrove species classification	5 species, 99.88%
3-D-2-D Hybrid Lightweight CNN (Hybrid-LtCNN) **[8]**	Custom hyperspectral canopy dataset	Plant species classification (canopy)	100 species, Macro F1 = 0.9935
Automatic fused multimodal deep learning **[9]**	Multimodal-PlantCLEF dataset	Plant identification	979 classes, 82.61%
SE-SK-CapResNet (Capsule + Residual + Attention) **[10]**	PlantVillage subset, AI Challenger 2018, Tomato Leaf Disease dataset	Plant leaf disease classification	6/6/2006 classes, 98.58% / 95.08% / 97.19%
EfficientNetB0 + LIME + XAI **[11]**	Kaggle New Plant Diseases dataset	Plant disease classification with XAI	38 classes, 99.69%

Methods Used	Dataset Type	Task	Classes & Accuracy
PSO Optimized Cascaded Network (ResNet50 + SVM) **[12]**	Review of 94 studies	Medicinal plant classification	N/A, 99.60%
Hoya-CNN2GNN (Knowledge Distillation) **[13]**	Smartphone-captured medicinal plant image dataset	Plant classification (leaf and flower)	7 classes, 97.16% / 95.18%
Review of deep learning methods **[14]**	Custom Hoya leaf and flower datasets	Review of plant disease detection	26/28 classes, N/A
TLMVIT: Transfer learning + ViT + MLP **[15]**	PlantVillage subset and Wheat Rust dataset	Plant disease classification	15/3 classes, 98.81% / 99.86%
PointNet, PointNet++, PointCNN, DGCNN **[16]**	Custom 3-D LIDAR sorghum point-cloud dataset	Plant organ segmentation (LIDAR)	3 classes, 91.50%
EfficientNetB3-AADL **[17]**	RGB plant disease image dataset	Multi-class plant disease classification	52 classes, 98.71%
Review of ML/DL methods & benchmarking **[18]**	Review literature + widely used plant disease datasets	Review of plant disease detection	N/A
Deep CNN + LBP feature fusion **[19]**	PlantVillage subsets: apple, tomato, grape leaves	Multi-class plant leaf disease classification	4/10/4 classes, 98.8% / 96.5% / 98.3%
PlantCLEF 2023 benchmark **[20]**	Global-scale benchmark dataset	Image-based plant identification	80,000 species, MA-MRR 0.64079
Inception-v4 and Inception-ResNet-v2 ensemble **[21]**	PlantCLEF 2023 dataset	Large-scale plant classification	80,000 species, MA-MRR 0.61813

Methods Used	Dataset Type	Task	Classes & Accuracy
Multi-organ deep learning with DC-GAN; Inception-ResNet-v2 **[22]**	PlantCLEF 2015 multi-organ dataset	Multi-organ plant species classification	1000 species, 86.00%
PlantCLEF 2022 benchmark **[23]**	Global-scale benchmark dataset	Image-based plant identification	80,000 species, MA-MRR 0.626
Evaluation of Flora Incognita app **[24]**	Database images + field observations	Plant image identification	542/280 species, 79.6% / 85.3%
Faster R-CNN **[25]**	Custom field image dataset	Plant and weed identification	2 categories, mAP ≈ 31%
Image processing + Random Forest **[26]**	Custom medicinal leaf image dataset	Medicinal plant identification	10 classes, 94.54%
Deep feature extraction (MobileNetV2, etc.) + SVM **[27]**	Custom Vietnamese environment image dataset	Plant species identification	109 species, 83.9%
Score-based fusion of multi-CNN **[28]**	Multi-organ plant image dataset from PlantCLEF 2015 + Internet	Multi-organ plant identification	50 species, 89.80%
Multi-Organ Integrated PlantNet (CNNs + SVM) **[29]**	Custom multi-organ plant image dataset	Plant identification	100 species, 86.60%
HGO-CNN and Multi-Scale HGO-CNN **[30]**	PlantCLEF 2015 multi-organ benchmark	Multi-organ plant classification	1000 species, Sobs 0.717 / Simg 0.690

2.4 Research Challenges

Even though deep learning has seen extraordinary success in plant identification, organ classification, disease detection and edibility prediction, multiple research challenges still exist. These issues are due to constraints in datasets, model architecture, deployment environments and interpretability which greatly hinder the DL systems in the domains ranging from agriculture and conservation through to digital botany [1].

2.4.1 Limited Availability of Annotated Datasets

Annotated datasets are also limited in scale and quality, for rare species or uncommon organs early-stage or rare disease conditions [1]. For example, data sets might contain thousands of images of leaves. However, very few images of flowers or fruits may be present. The compelling number of annotations leads to the development of robust models with limited generalization capability for unseen species or organs. Manual annotation is costly and time-consuming, as it requires expert botanical knowledge to ensure accurate labeling [1].

2.4.2 Class Imbalance

The datasets still might follow long-tail distributions where common organs or diseases are overrepresented, and rare organ early staged diseases are underrepresented [14]. Consequently, when the classes are proportioned, there is an internal tendency of the models to over-fit possible discussion of. This favors accuracy for frequent classes but leads to low recall among less common organs. In disease detection, this can lead the models to miss out on early warning signs that are important for intervention. To mitigate these effects, techniques such as oversampling, class-aware loss functions and synthetic data generation are used [18].

2.4.3 Domain Shift and Poor Generalization

These solutions are known to tackle the challenges associated with domain shift from one geographical location or season to another, involved in both the acquisition of computer vision videos as well as their projection onto various devices [7].

Often models which have been trained in sterile lab conditions fail to translate due to differences in perceptual characteristics on ground, including lighting, background, occlusion, motion blur or environmental noise [28] for instance, a CNN trained on isolated high-resolution images of leaves may fail to accurately classify cluttered canopy leaves.

2.4.4 Subtle and Complex Organ and Disease Patterns

One of the challenges is identifying small differences between organs, or patterns of early-stage disease. Fine visual details such as hairline lesions, partial discoloration, overlapping leaves and small fruits make it challenging for models to learn in detail [4].

They have shown that misdiagnosing them negatively impacts the accuracy of early disease detection methods and lowers the reliability of multi-organ classification systems. Architectures like attention-augmented CNNs, ViTs, and hybrid CNN-GNN models lead to higher sensitivity for subtle patterns [13].

2.4.5 Evaluation Inconsistency

Studies have not used a consistent set of evaluation metrics, dataset splits or benchmark protocols [14]. However, accuracy, F1-score, precision and recall are reported differently by different research groups leading to a poor comparison of the models. This inconsistency in evaluation hampers reproducibility and can also result in over-optimistic estimates of model performance when deployed to the real world [14].

2.4.6 Multimodal Integration Constraints

The inclusion of multi-modality data (RGB images, LiDAR, hyperspectral, and 3D imaging) can greatly enhance organ detection, disease classification and identification [8]. The conversion of humans into pixels is not a one-to-one mapping and there are various modalities that need to be aligned, managed across different resolutions for downstream analysis to deal with large multimodal data streams requiring sophisticated computational pipelines. Training multimodal models is complex and incurs higher inference costs as the real-time deployment of SIP (Speech, Image, Text) integrated classifiers is difficult [8].

2.5 Conclusion

Deep learning has come to revolutionize automated plant identification. Moreover, CNN-based models produce hierarchical features while adopting pre-trained networks (EfficientNet, ResNet and DenseNet) for transfer learning can yield significant classification accuracy on multi-organ and dynamic species datasets (92–94%) [3-4]. The foundation of botanical taxonomy was based on morphological analysis of leaves, flowers, fruits, seeds and roots [5]. Global contextual dependencies are captured through partial or non-overlapping pixel cropping with vision transformers, improving subtle organ and early disease detection which has shown extraction accuracies of 94% [6].

Improved disease detection and plant phenotyping have followed from attention-augmented CNNs, hybrid CNN-GNN architectures, and multimodal imaging (LiDAR, hyperspectral). Through this and the importance of accurate orientation, those techniques allow for precise organ segmentation, subtle lesion localization, and early-stage disease detection at more than

91% F1-scores [8-14]. Multi-task learning schemes further improve efficiency and consistency by predicting species, organ type, health status, and edibility simultaneously [15].

Early digital and machine learning approaches tried to automate feature extraction with handcrafted characteristics that achieved a moderate accuracy between 64% and 92%, however they were very sensitive to environmental variability, and insufficient for multi-task applications [26], but traditional methods were not scalable to large datasets or multiple organ types and could only be performed by highly experienced botanists due to the labor intensity required [28].

CHAPTER 3
METHODOLOGY

3.1 Introduction

The methodology presented outlines the combined design and execution of a DL-based multi-classification plant system using raw images to determine five primary characteristics: plant name, species, organ, health status, and edibility. In contrast to single-task models, which are limited to only one objective, this pipeline allows multiple tasks to be accomplished simultaneously, providing for a comprehensive analysis of the plant.

Another difficulty involved in identifying plants in a natural environment relates to the variability of light conditions, background complexity, stages of growth, and subtle signs of disease. To alleviate these issues, the approach includes preprocessing the dataset to eliminate artifacts; developing robust features; and enhancing model generalization and reliability.

Because the multi-classification approach depends on both local and global image features, the methodology includes a combination of preprocessing steps, transfer learning, and optimization of architectures to facilitate balanced training of the different components throughout the entire pipeline of this multi-functional plant classification algorithm. Also, each component of the methodology is documented in a way to ensure that reproduction and interpretation can occur easily. Finally, a comparative evaluation was performed to determine the value of using ResNet50, EfficientNetB3, EfficientNetV2L and ConvNeXtBase as pre-trained models for implementing the same comparative analysis protocol on the same plant image classification database for comparing performance and finding the best model to execute this type of task.

3.2 Dataset Description

This custom dataset consists of a multi-class deep learning classification system to classify plants with attributes such as name, species, type of organ (root, leaf, flower, fruit, seed), health status and whether they are edible. The images used to create this dataset are obtained through various publicly available online repositories as well as Kaggle datasets (such as flower classification, sunflower leaves, root images, rose leaves, national flower, plant diseases, flower-299 and fruit classification) and cover a total of 262 classes comprising 90,970 images from 26 different plant types; therefore, constituting a very large-scale and diverse corpus of images to use for training and testing purposes. The images in this dataset exhibit variability in real-world conditions consequently, allowing for robust generalization and minimizing the chances of overfitting to perfect representation of plants contained within the dataset.

To standardize the images, several different preprocessing techniques were employed (resizing images to the input dimensions of the model, retaining all colors in the processed images). Data augmentation techniques (rotation, flipping, zoom, shift and adjusting brightness) were also employed to improve the diversity of the images within the dataset and reduce opportunities for overfitting. The final augmented dataset was split into a stratified training (80%) and validation (20%) sets to ensure class distributions are reflected in the split dataset.

3.3 Proposed Methodology

This methodology provides a single, unified multi-task deep-learning (DL) system that utilizes one input image of a plant and predicts the plant's five attributes: common name, species, organ, health status, and edible or inedible through a series of shared feature representations. Because the modeling process is performed through a single pipeline for all five attributes, it does not require separate models for each task (attribute) and, therefore, increases prediction consistency and efficiency through the knowledge gained from learning similarities.

3.3.1 Dataset Preparation: Raw images are curated by performing quality checks, removing duplicates, resizing, normalizing, and augmenting to make picture data maintain environmentally significant variability while having standardized input. Additionally, each image sample is tagged with a multi-label encoding for every task that is to be predicted.

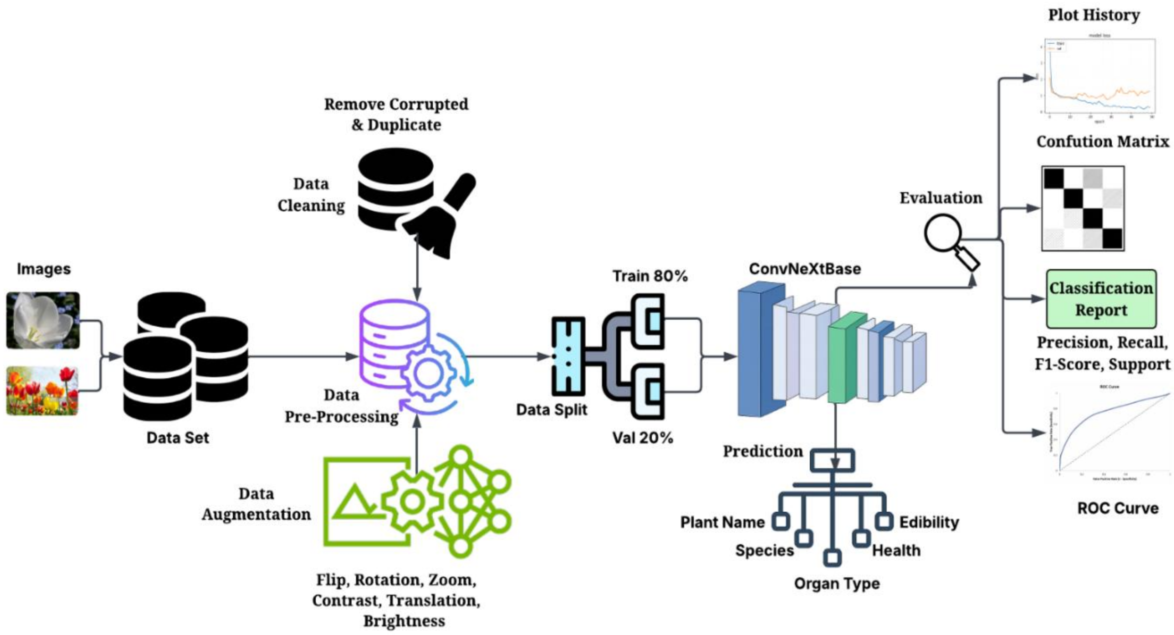
3.3.2 Feature Extraction & Fine-tuning: For feature extraction, the use of transfer learning (pretrained backbone) was performed on the following four (4) architectures: ResNet50, EfficientNetB3, EfficientNetV2L, and ConvNeXtBase, for robust initial feature extraction with the feature extraction process for plants. Following transfer learning extraction with pretrained backbone networks, fine-tuning was performed on the adopted training dataset for the specific domain (plant) for domain-specific feature extraction.

3.3.3 Model Training & Optimization: Supervised training was accomplished through an adaptive optimization solution, and parameters tied to the model training process were augmented with the use of an early stopping strategy, checkpointing, and learning rate scheduler to mitigate overfitting and ensure convergence stability.

3.3.4 Evaluation & Model Selection: Model performance was quantified based on the following five (5) metrics: (i) accuracy, (ii) precision, (iii) recall, (iv) F1 score, and (v) confusion matrices by task. Based on the results of the comparative analysis, ConvNeXtBase produced the best overall results relative to all objectives evaluated.

The completed pipeline integrates all phases of the plant image analysis process, from dataset preparation, to transfer learning, to multi-task learning, to thorough evaluation to provide the user with a reliable, scalable means of performing plant image analysis.

Fig 3.3.1: Proposed Methodology



3.4 Conclusion

The plant identification is a multi-task dimensional deep learning system (DL) capable of identifying the name, species of plant, parts of plant (organ), condition of plant (health), and edibility of plants from a single image of that plant. The systematic multi-task dimensional DL Pipeline incorporated into the classification system includes image acquisition; dataset curation; multi-label encoding; transfer learning; feature extraction; fine-tuning; validation-based optimization; and rigorous evaluation, providing a robust basis for making reliable predictions. A unique contribution of this work is developing a controlled comparison of the four pretrained models (ResNet50, EfficientNetB3, EfficientNetV2L, ConvNeXtBase) used to generate empirical evidence on their usefulness in identifying plants. The multi-task dimensional DL system creates a common platform to make multi-dimensional inferences from a single image, compared to traditional siloed systems.

Chapter 4
Result Analysis and Discussion

4.1 Introduction

This chapter provides experimental results and analyses for the proposed plant identification system, which employs deep learning techniques. The primary purpose of this chapter is to assess the performance of the ConvNeXtBase model, which serves as the baseline model for plant classification. To assess the performance of the ConvNeXtBase model and validate its effectiveness as a plant identification and classification system, we conduct a comparative performance evaluation with multiple other models using deep learning techniques. The models generated multiple performance metrics (accuracy, precision, recall, and F1 score), which help evaluate the performance of the models. Additionally, graphical visualization tools (accuracy and loss curves, roc curves, and confusion matrices) assist in assessing the performance of each of the models on plant identification and classification tasks. The results of the model evaluations yield important information about the strengths and weaknesses of the individual models and demonstrate the effectiveness of the ConvNeXtBase model in plant identification and classification tasks.

4.2 Result Analysis

This section presents an analysis of the various experimental outputs generated by the evaluation of the deep learning models built to classify various types of plants. Model was evaluated using standard classification metrics as well as through several visualization methods for each Classifier to ascertain the ability of the model to accurately identify different types of plants based upon images. Also included in the analyses of results is whether the proposed deep learning method will be able to effectively deal with the vast amount of data generated by multiple images of plants taken under differing conditions, while still providing accurate results for each method.

4.2.1 Performance Evaluation Metrics

The performance of the models is evaluated using various standard metrics. The standard metrics provide a quantitative measure of how well the models perform on classification tasks.

Accuracy: Accuracy measures the overall correct predictions of how well the model predicted the plant classes. The accuracy can be obtained by dividing the number of correctly predicted samples over the total samples.

$$\text{Accuracy} = \frac{\mathbf{TP + TN}}{\mathbf{TP + FP + TN + FN}} \tag{1}$$

Precision: Precision measures the number of correctly predicted positive predictions compared to all the positive predictions.

$$\text{Precision} = \frac{\mathbf{TP}}{\mathbf{TP + FP}} \tag{2}$$

Recall: Recall measures how well the model can identify all relevant samples in the dataset.

$$\text{Recall} = \frac{\text{TP}}{\text{TP + FN}} \tag{3}$$

F1-Score: The F1-Score is the harmonic mean (average) between precision and recall. The F1-Score provides a balanced representation of the model performance, especially on an imbalanced dataset.

$$\text{F1-Score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{4}$$

4.3 Performance Comparison of Four Architectures

The four deep learning models, ResNet50, EfficientNetB3, EfficientNetV2L, and ConvNeXtBase, were evaluated on 5 plant identification tasks including Plant Name, Plant Species, Plant Organ, Plant Health, and Plant Edibility. In terms of training accuracy, ResNet50 scored between 94.77-97.91% and for validation accuracy, it was between 91.73-93.95%. It performed well on general classification tasks such as Plant Name or Plant Species, but because it lacks the ability to extract fine features from the images, it had lower performance for classification of Plant Organ and Plant Health. The use of EfficientNetB3 helped improve overall performance, the training and validation accuracies were 95.85-97.80% and 93.78-96.67%, respectively. In addition to achieving a strong F1-score, precision, and recall for both Plant Species and Edibility; it also scored high with F1-score, precision, and recall for all other plant identification tasks. In comparison to EfficientNetB3, EfficientNetV2L demonstrated better training and validation accuracies (97.41-99.72% and 94.17-97.53%, respectively) by providing excellent feature extraction of fine detail patterns in the images but did show slight decreases for the Plant Health and Plant Organ classification tasks. ConvNeXtBase consistently outperformed all the other models with training accuracies ranging from 96.50-98.93% and validation accuracies ranging from 94.96-98.57%, and it achieved. F1-score, precision, and recall scores that were both high and stable across all 5 identification tasks. It can capture very subtle visual differences, making it very good at complex problems in classifying and identifying plant health. Overall, ConvNeXtBase is the most accurate and robust plant identification model, while EfficientNetV2L and EfficientNetB3 provide good alternatives, and ResNet50 works well on less complex identification tasks.

Table 4.3.1: Comparison of Different Models

Model	**Task**	**Train Acc**	**Val Acc**	**F1-score**	**Precision**	**Recall**
ResNet50	*Plant Name*	*97.91%*	*93.95%*	*94%*	*94%*	*93%*
	Plant Species	*95.46%*	*93.21%*	*93%*	*94%*	*92%*
	Plant Organ	*97.78%*	*93.82%*	*94%*	*80%*	*79%*
	Plant Health	*95.70%*	*91.73%*	*92%*	*92%*	*90%*
	Plant Edibility	*94.77%*	*92.47%*	*93%*	*92%*	*90%*
EfficientNetB3	*Plant Name*	*96.53%*	*95.45%*	*97%*	*97%*	*97%*
	Plant Species	*97.40%*	*96.80%*	*98%*	*98%*	*98%*
	Plant Organ	*96.40%*	*95.27%*	*97%*	*97%*	*97%*
	Plant Health	*95.85%*	*93.78%*	*94%*	*94%*	*94%*
	Plant Edibility	*97.80%*	*96.67%*	*98%*	*98%*	*98%*
EfficientNetV2L	*Plant Name*	*99.53%*	*96.92%*	*96%*	*96%*	*96%*
	Plant Species	*99.35%*	*97.53%*	*97%*	*97%*	*97%*
	Plant Organ	*99.50%*	*97.25%*	*95%*	*95%*	*96%*
	Plant Health	*97.41%*	*94.17%*	*95%*	*95%*	*96%*
	Plant Edibility	*99.72%*	*97.50%*	*97%*	*97%*	*97%*
ConvNeXtBase	***Plant Name***	***98.63%***	***98.49%***	***98%***	***98%***	***98%***
	Plant Species	***98.75%***	***98.46%***	***98%***	***98%***	***98%***
	Plant Organ	***98.93%***	***98.57%***	***98%***	***98%***	***98%***
	Plant Health	***96.50%***	***94.96%***	***95%***	***95%***	***95%***
	Plant Edibility	***98.35%***	***98.10%***	***98%***	***98%***	***98%***

4.4 Model Output Visualization

Visualization techniques help analysts to better comprehend the models' performance when trained and evaluated. Visual representations will help to illustrate how the models are learning and how their classifications are behaving.

4.4.1 Accuracy and Loss Comparison

The accuracy and loss graphs provide an assessment of model performance during training and validation. They show how the model has learned from the data throughout the course of multiple epochs of training. The training accuracy improves steadily with each epoch of training while the training loss reduces steadily over time. This demonstrates that the model learned meaningful relationships in the data. The validation accuracy and loss curves show similar stable behavior during learning with no apparent significant overfitting; this provides evidence that the training strategy employed for this model was effective.

Fig 4.4.1: Accuracy Plot for Plant Name

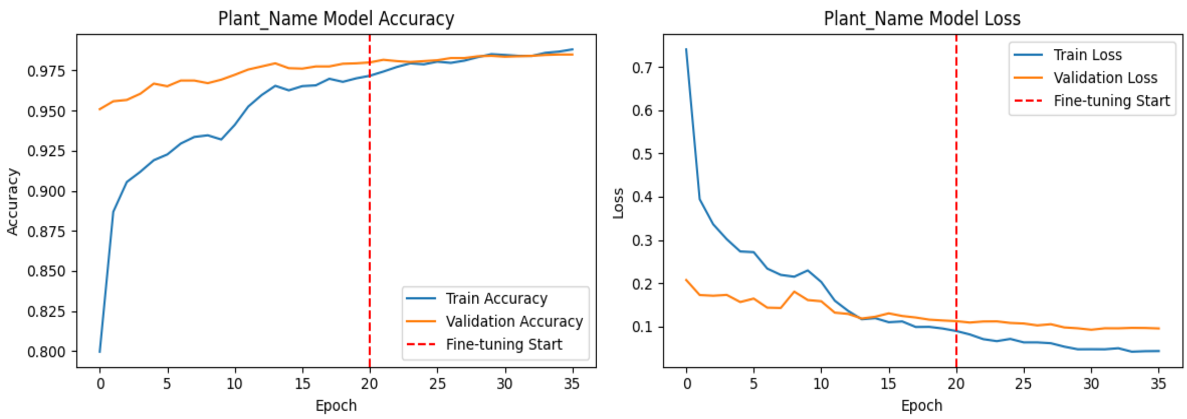


Fig 4.4.2: Accuracy Plot for Plant Species

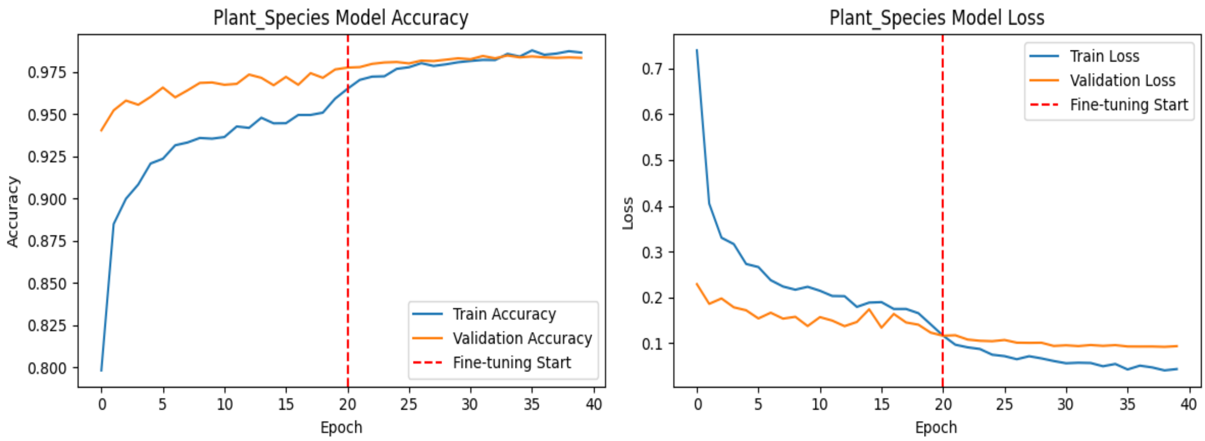


Fig 4.4.3: Accuracy Plot for Plant Organ

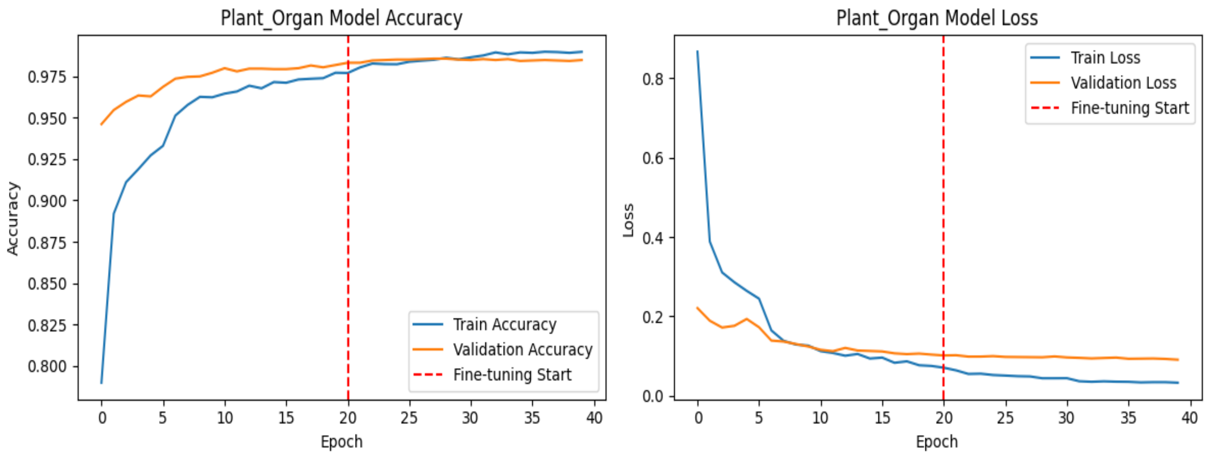


Fig 4.4.4: Accuracy Plot for Plant Health

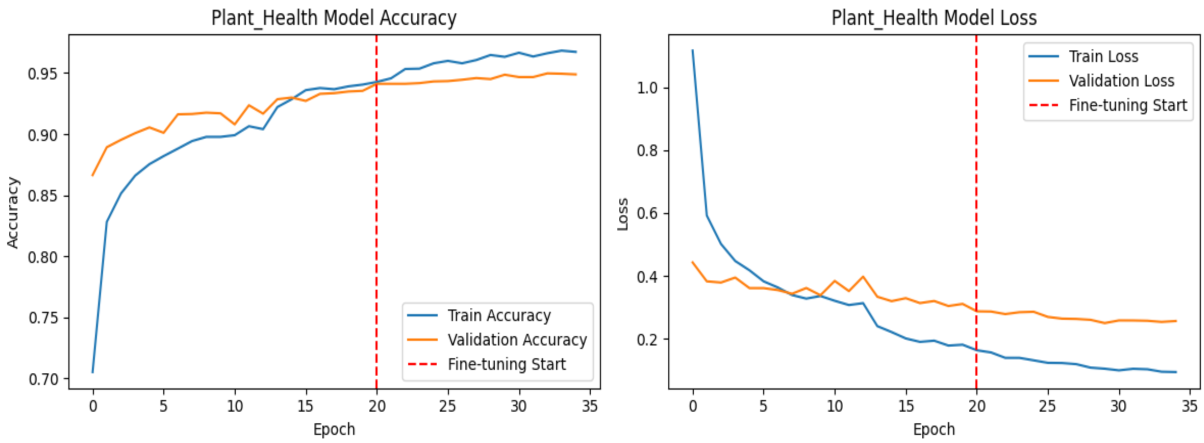
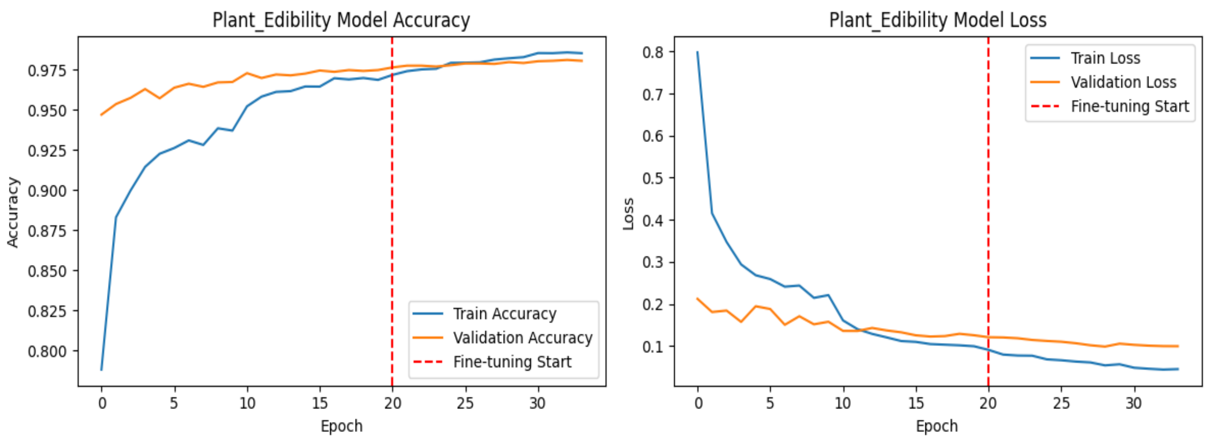


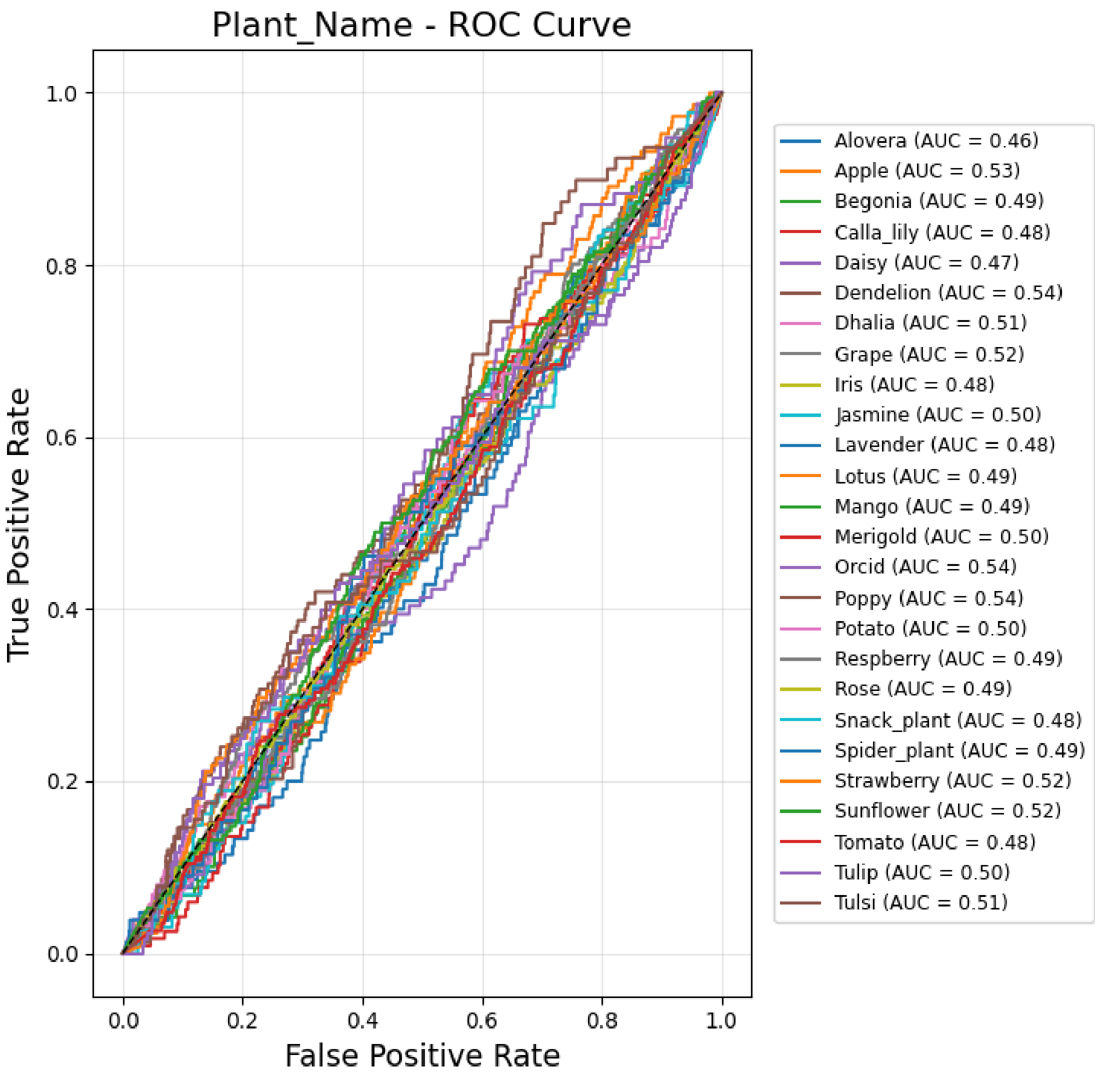
Fig 4.4.5: Accuracy Plot for Plant Edibility



4.4.2 ROC Curve

The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (ROC) curve is a useful tool that allows researchers to assess how well their models classify across various threshold settings. By plotting the FPR, it is possible to determine how well a model can classify observations compared to other models. The closer the ROC curve of a particular model is to the upper left corner, the higher the classification ability of that model.

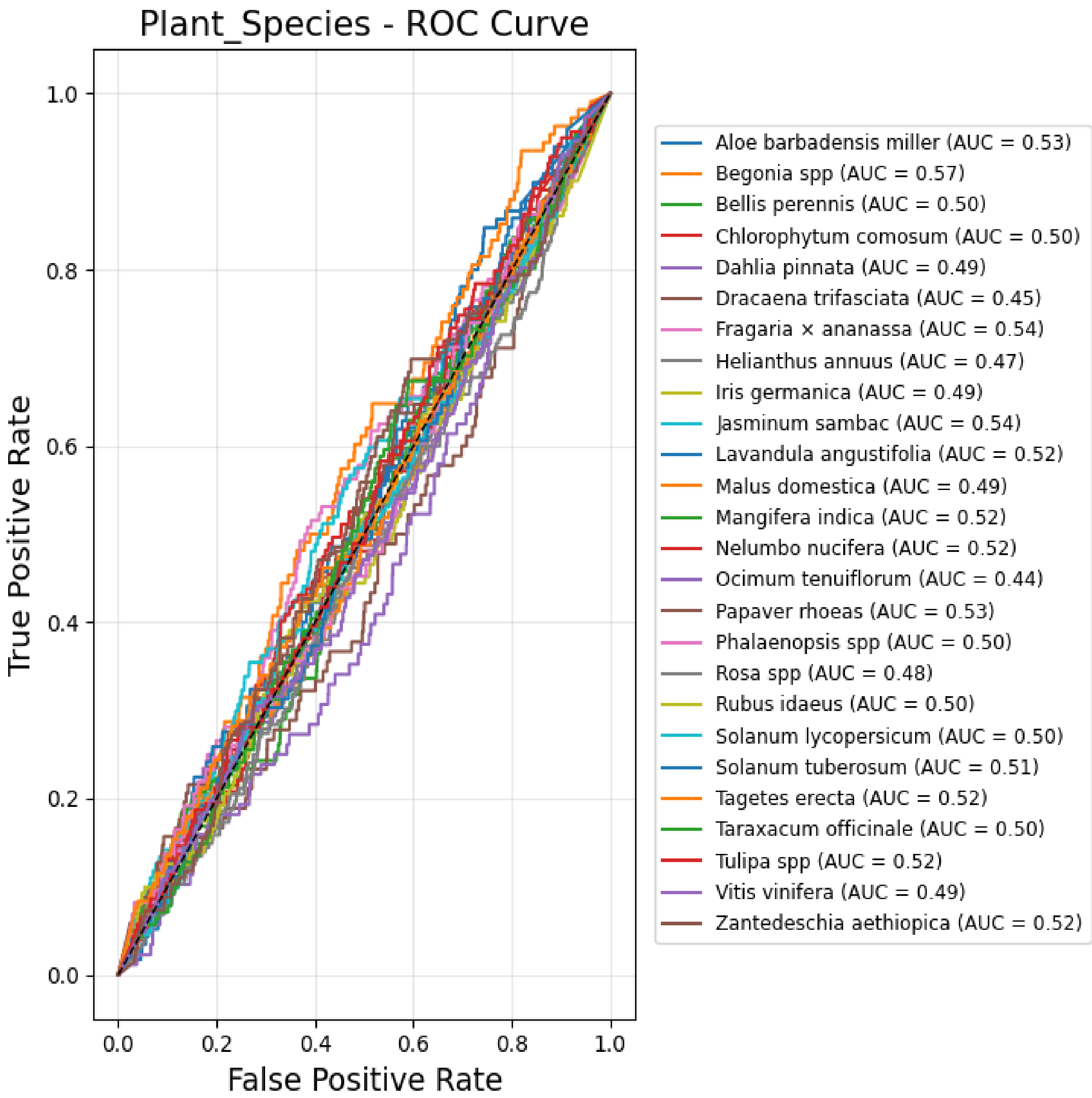
Fig 4.4.2.1: ROC Curve for Plant Name



This is an ROC curve titled "Plant Name ROC Curve" showing the performance of a plant classification model.

The False Positive Rate is plotted on the x-axis, and the True Positive Rate is on the y-axis. All individual plant curves hover closely around the diagonal black dashed line (AUC= 0.50), meaning the model performs roughly the same as random guessing. Among the listed categories, Dandelion, Orcid, and Poppy achieved the highest Area Under the Curve (AUC = 0.54), while Alovera had the lowest (AUC = 0.46).

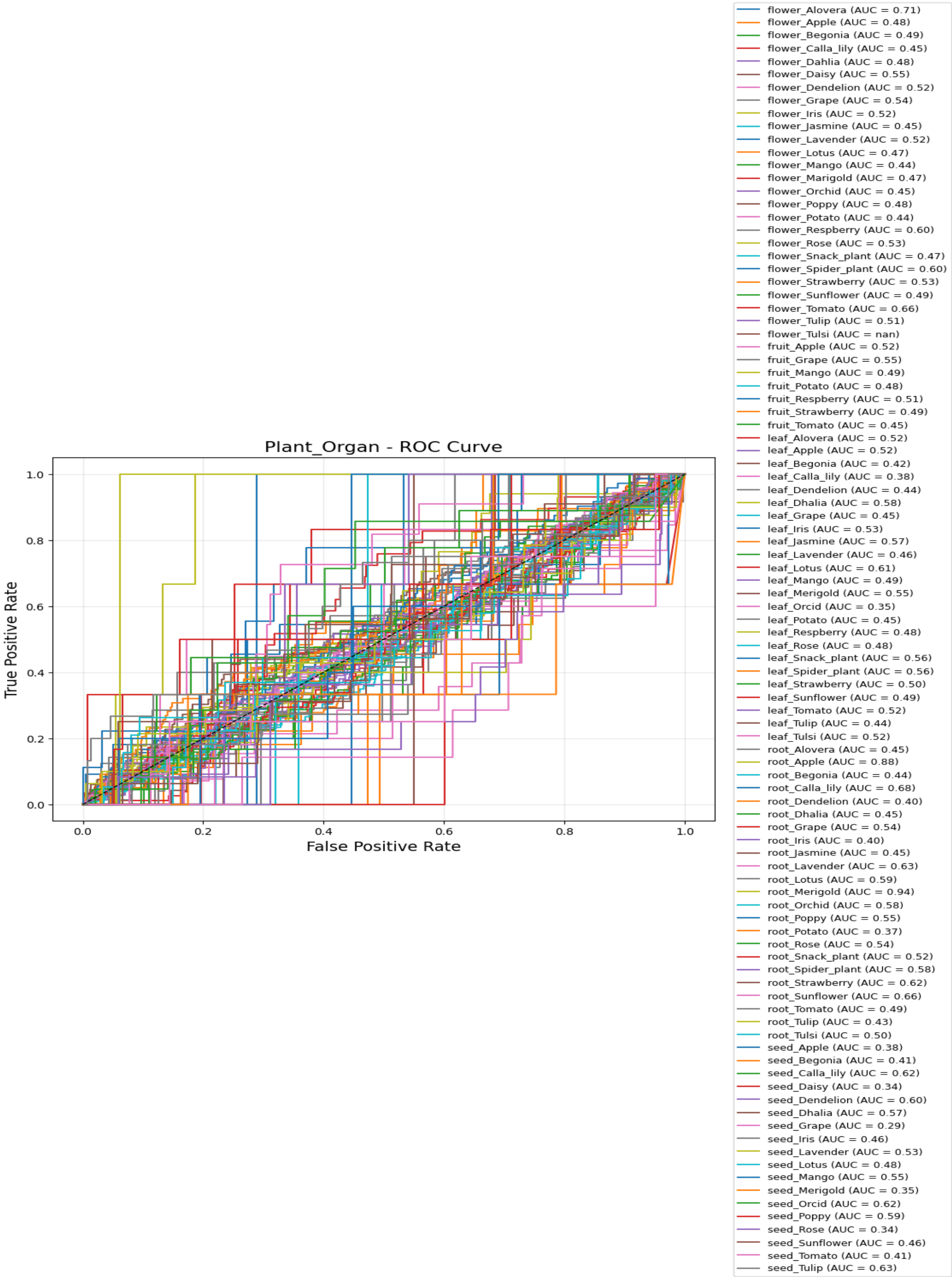
Fig 4.4.2.2: ROC Curve for Plant Species



This ROC curve, titled "Plant Species ROC Curve," evaluates a classification model using scientific names.

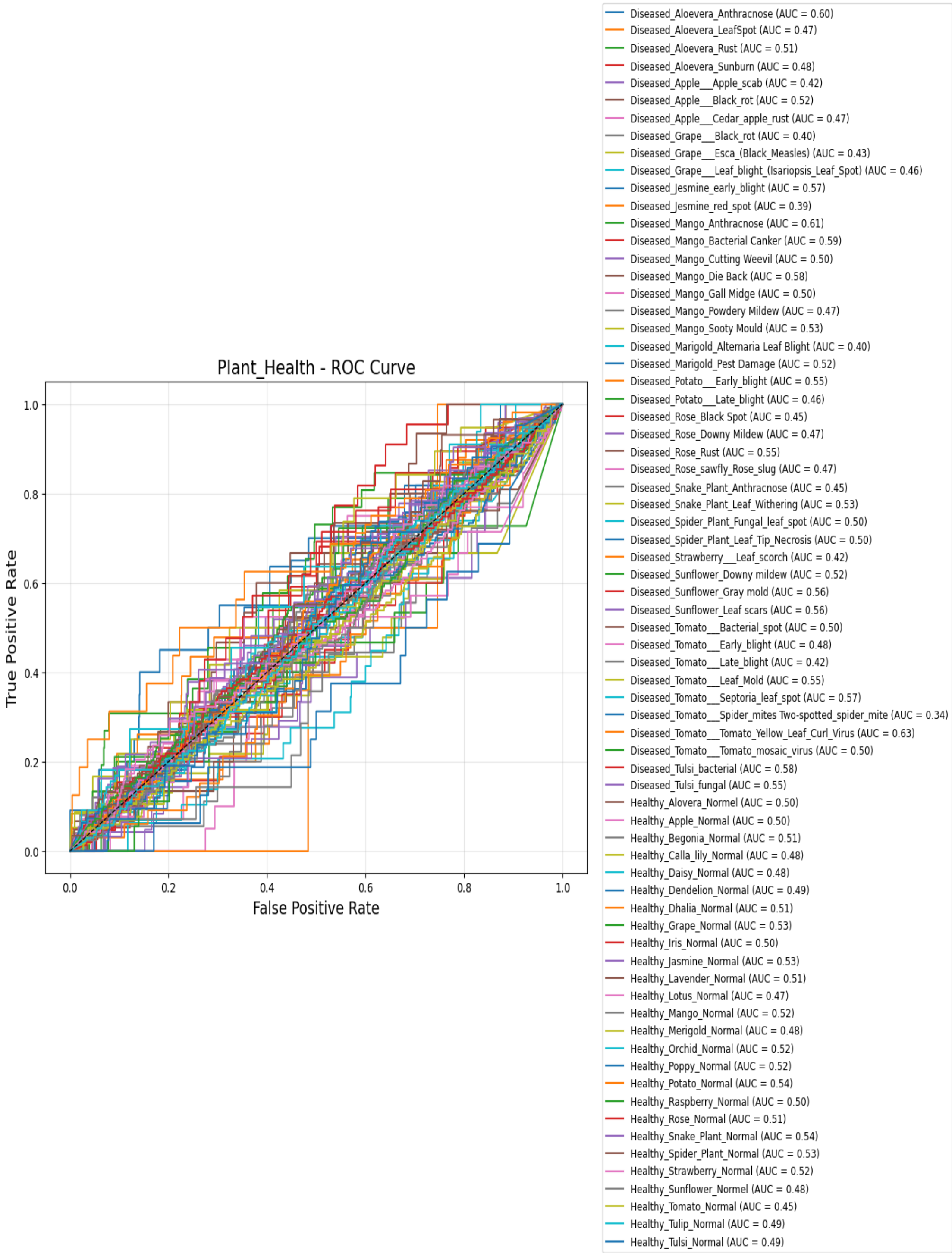
The individual species curves closely follow the diagonal random-chance line, indicating the model performs near random guessing overall. Begonia spp achieved the highest performance with an AUC of 0.57. Conversely, Ocimum tenuiflorum showed the lowest performance with an AUC of 0.44.

Fig 4.4.2.3: ROC Curve for Plant Organ



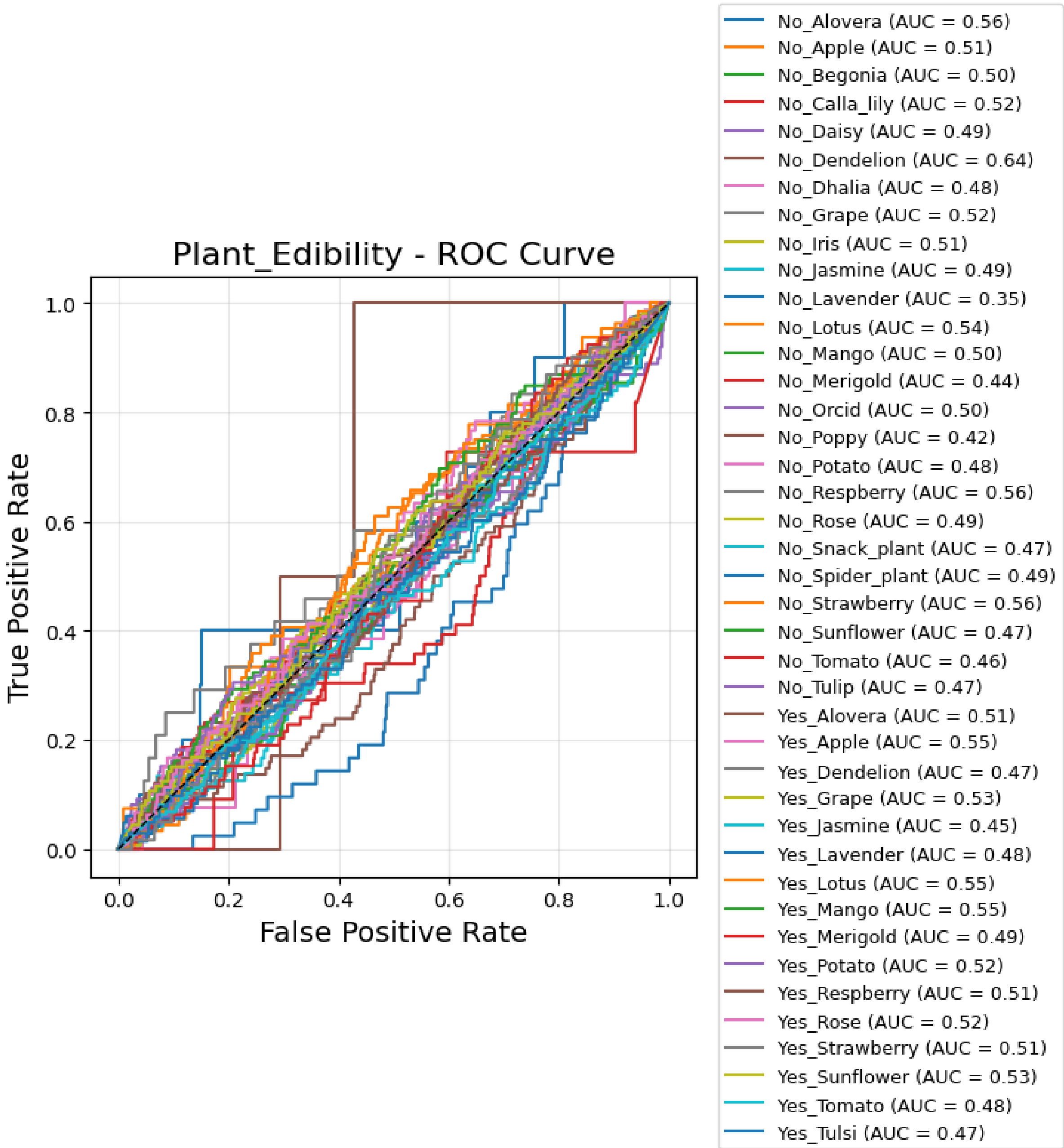
This ROC curve, titled "Plant Organ ROC Curve," evaluates classification performance across specific plant organs (flowers, fruits, leaves, roots, seeds). Performance varies significantly compared to the previous models. root_Merigold achieved the highest performance with an exceptional AUC of 0.94, followed closely by root_Apple at 0.88.

Fig 4.4.2.4: ROC Curve for Plant Health



This ROC curve, titled "Plant Health ROC Curve," tracks classification performance for both diseased and healthy plant conditions. Most categories closely cluster along the diagonal baseline. Diseased_Tomato_Tomato_Yellow_Leaf_Curl_Virus achieved the highest classification performance with an AUC of 0.63.

Fig 4.4.2.5: ROC curve for Plant Edibility



This ROC curve, titled "Plant Edibility ROC Curve," evaluates classification performance based on whether parts of the plants are edible ("Yes") or not ("No"). While most categories group closely along the diagonal random line, a few distinct deviations occur. No_Dendelion achieved the highest performance with an AUC of 0.64.

4.4.3 Overall Performance Analysis

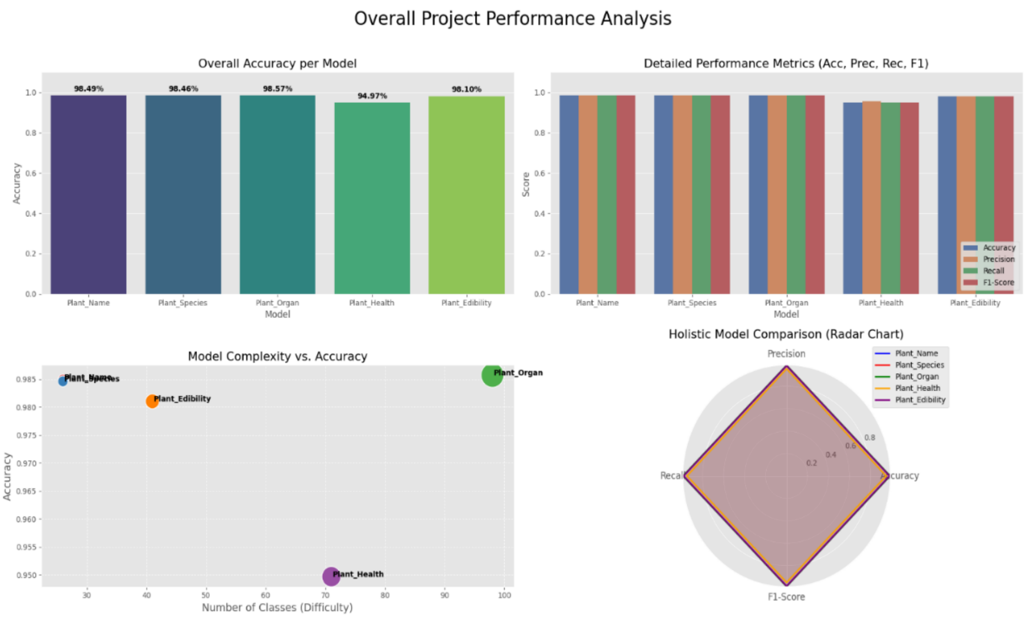
Overall Performance Evaluating how well the models have performed involves combining both numerical and graphical measures so that we can determine the effectiveness of the given architecture. The results reveal that on almost all different types of measures that were taken the ConvNeXtBase architecture performed higher than the other models used for comparison.

Table 4.4.3.1: Model Performance Metrics

Model	Classes	Accuracy	Precision	Recall	F1-Score
Plant Name	26	0.9849	0.9850	0.9849	0.9849
Plant Species	26	0.9846	0.9848	0.9846	0.9846
Plant Organ	98	0.9857	0.9849	0.9857	0.9845
Plant Health	71	0.9497	0.9549	0.9497	0.9496
Plant Edibility	41	0.9810	0.9814	0.9810	0.9810

This table evaluates the performance of five distinct classification models: Plant Name, Plant Species, Plant Organ, Plant Health, and Plant Edibility. Each model is assessed across varying class counts using four key metrics: Accuracy, Precision, Recall, and F1-Score. Overall, the models demonstrate exceptional predictive capability. The Plant Name, Plant Species, Plant Organ, and Plant Edibility models all achieve outstanding results, with scores consistently exceeding 98% across all evaluation criteria. While the Plant Health model shows slightly lower metrics, it still maintains high robust performance with scores hovering around 95%.

Fig 4.4.3.1: Overall Performance Analysis



This figure provides a comprehensive visual analysis of the project's model performance across four different charts. The top-left bar chart highlights the Overall Accuracy per Model, showing that Plant Organ achieved the highest accuracy at 98.57%, while Plant Health was the lowest at 94.97%. The top-right clustered bar chart breaks down the Detailed Performance Metrics, demonstrating strong consistency among Accuracy, Precision, Recall, and F1-Score for each model.

In the bottom-left, the Model Complexity vs. Accuracy scatter plot maps accuracy against the number of classes, revealing that the Plant Organ model handled the highest complexity while maintaining excellent accuracy. Finally, the bottom-right Radar Chart offers a holistic comparison, showing near-perfect, overlapping shapes that underscore the balanced and high-level performance across all evaluated metrics.

4.5 Detailed Analysis of Model Performance

The model uses ConvNeXtBase as the primary feature extractor backbone, which learns significant visual patterns from plant leaf images. The Global Average Pooling layer compresses the feature map into a concise feature vector, making it more efficient and less prone to overfitting. The Batch Normalization layer improves training stability, and the Dense layer with 512 units learns abstract representations. The Dropout layer mitigates overfitting by randomly setting neurons to zero during training. Finally, the output dense layer with 26 units conducts multi-class classification to determine various plant health statuses.

Table 4.5.1: Internal Architecture of Proposed Method

Layer (type)	**Output Shape**	**Param #**
input_layer_6 (InputLayer)	(None,320,320,3)	0
sequential (Sequential)	(None,320,320,3)	0
convnext_base (Functional)	(None,10,10,1024)	87,566,464
global_average_pooling2d (GlobalAveragePooling2D)	(None,1024)	0
batch_normalization (BatchNormalization)	(None,1024)	4,096
dense (Dense)	(None,512)	524,800
dropout (Dropout)	(None,512)	0
dense_1 (Dense)	(None,26)	13,338

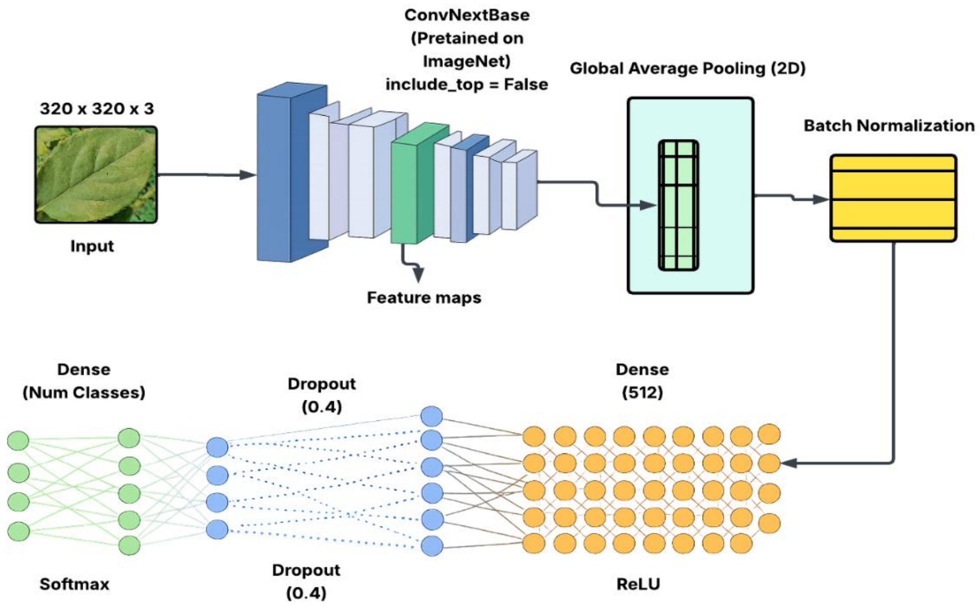
This table outlines the internal architecture of the proposed deep learning model. It processes 320 X 320 RGB images through an input layer and a sequential preprocessing framework. The core feature extraction is handled by a heavy convnext_base functional layer, which reduces the spatial dimensions to 10 X 10 while outputting 1024 channels, accounting for the vast majority of the network's parameters (87,566,464).

Following feature extraction, a GlobalAveragePooling2D layer flattens the spatial dimensions into a 1024-element vector. This vector passes through a BatchNormalization layer (4,096 parameters) and a fully connected Dense layer with 512 units (524,800 parameters). A Dropout layer is applied next to prevent overfitting. Finally, a second Dense output layer maps the features down to 26 classes (13,338 parameters) for the final classification task.

Parameter Summary

- **Total parameters:** 88,108,698 (336.11 MB)
- **Trainable parameters:** 540,186 (2.06 MB)
- **Non-trainable parameters:** 87,568,512 (334.05 MB)

Fig. 4.5.1: Proposed Method Architecture



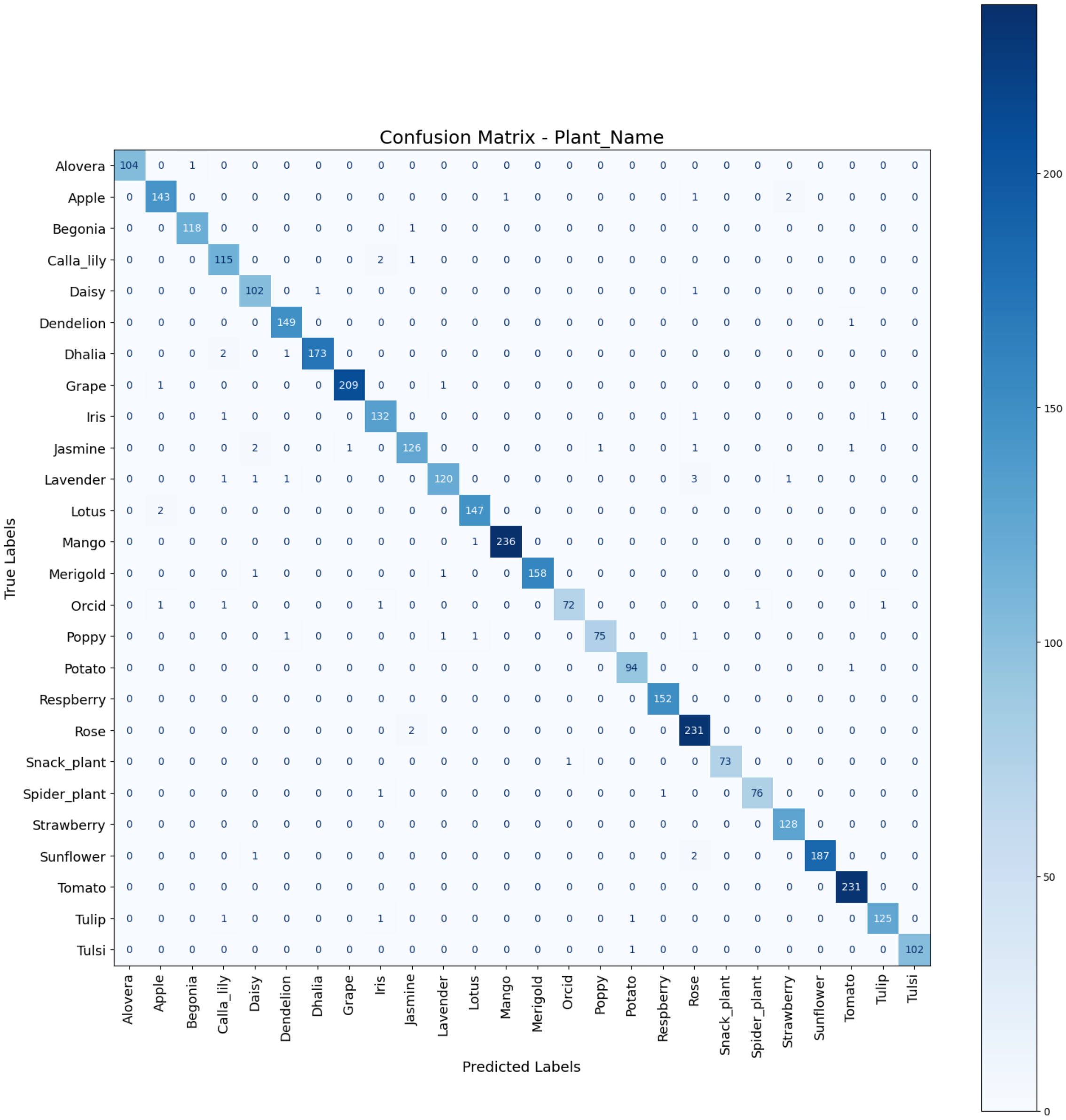
This Figure illustrates the architecture of the proposed deep learning framework for plant multi-classification. The model takes a 320 × 320 × 3 RGB image as input, which is preprocessed and normalized before being passed to a pretrained ConvNeXtBase backbone (include_top=False) for deep feature extraction. The backbone learns hierarchical features ranging from low-level patterns, such as edges and textures, to high-level features including leaf shape, color, venation, and disease characteristics.

The extracted feature maps are converted into a compact 1024-dimensional feature vector using Global Average Pooling (2D), reducing the number of trainable parameters and improving generalization. The feature vector is then normalized using Batch Normalization to stabilize training and accelerate convergence. Next, the normalized features are passed through a Dense layer with 512 neurons and a ReLU activation function to learn discriminative feature representations. A Dropout layer (0.4) is applied to reduce overfitting by randomly disabling 40% of the neurons during training. Finally, a Dense output layer with a Softmax activation function produces probability scores for the target classes. This architecture enables accurate and efficient classification of plant name, species, organ type, health status, and edibility while maintaining strong generalization performance.

4.5.1 Confusion Matrix Analysis

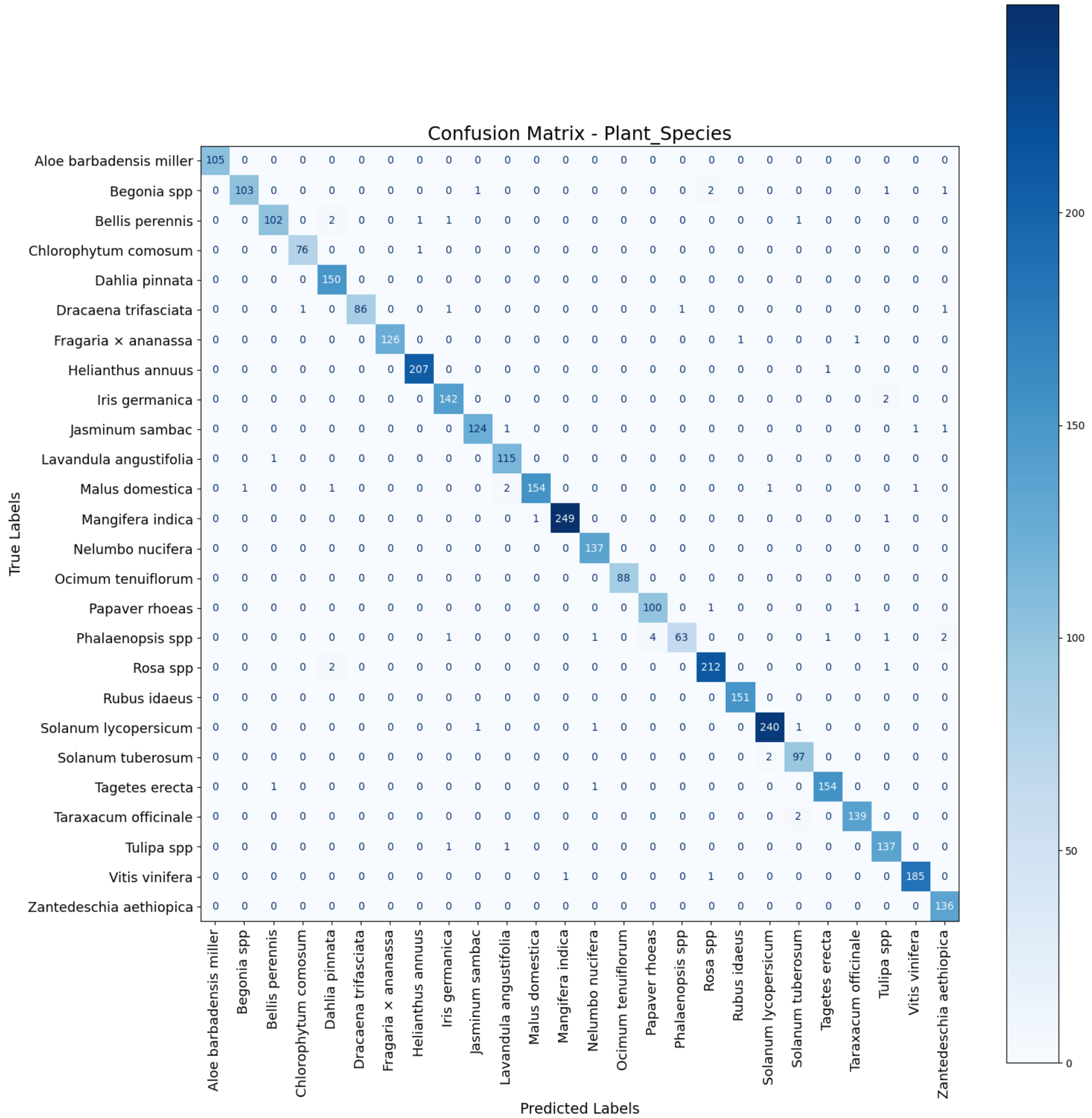
The classification performance of the model can be analyzed using a confusion matrix, which provides a graphical comparison between the actual and predicted class labels. Each row represents the actual class, while each column represents the predicted class. Correct classifications are located along the diagonal of the confusion matrix, whereas off-diagonal values indicate misclassifications. Analysis of the ConvNeXtBase confusion matrix shows that most samples were correctly classified, with predictions highly concentrated along the diagonal. Only a small number of samples were misclassified, primarily among classes with similar visual characteristics, such as closely related plant species, organs, or disease symptoms. These minor errors are likely caused by similarities in color, texture, and shape across certain categories. Overall, the confusion matrix demonstrates the strong classification capability of the ConvNeXtBase model, indicating high accuracy, robust feature extraction, and reliable performance in identifying plant names, species, organs, health status, and edibility from images.

Fig 4.5.1.1: Confusion Matrix for Plant Name



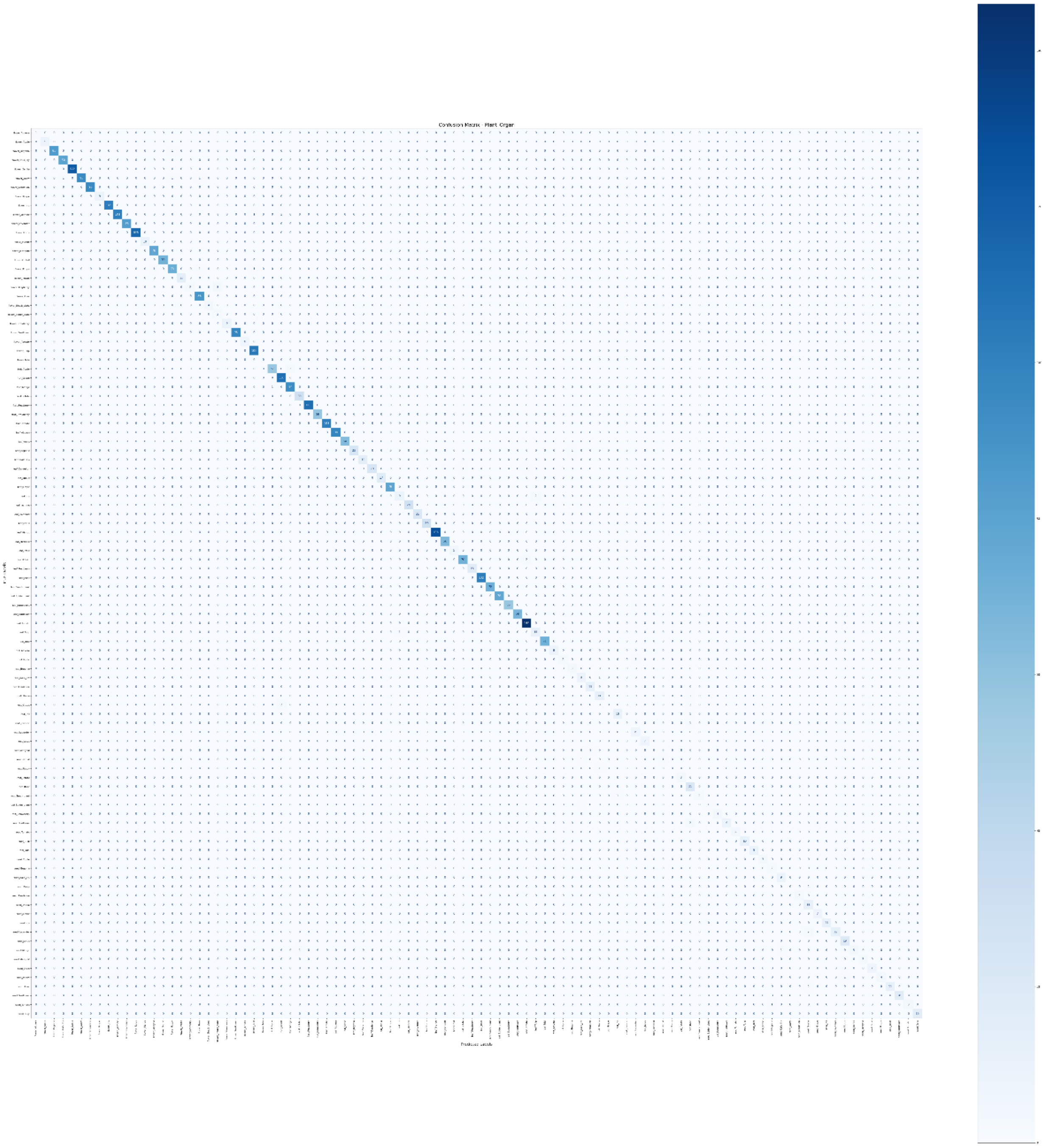
This figure displays the Confusion Matrix for the Plant Name model, evaluating its classification performance across 26 distinct plant categories. The prominent, dark blue diagonal line running from the top-left to the bottom-right confirms that the model is highly accurate, correctly predicting most instances for every class. Off-diagonal numbers represent misclassifications, which are exceptionally rare and scattered, appearing mostly as isolated single digits. The right-side color bar indicates the density of true labels, with categories like Mango, Rose, and Tomato showing the highest volume of correctly classified samples.

Fig 4.5.1.2: Confusion Matrix for Plant Species



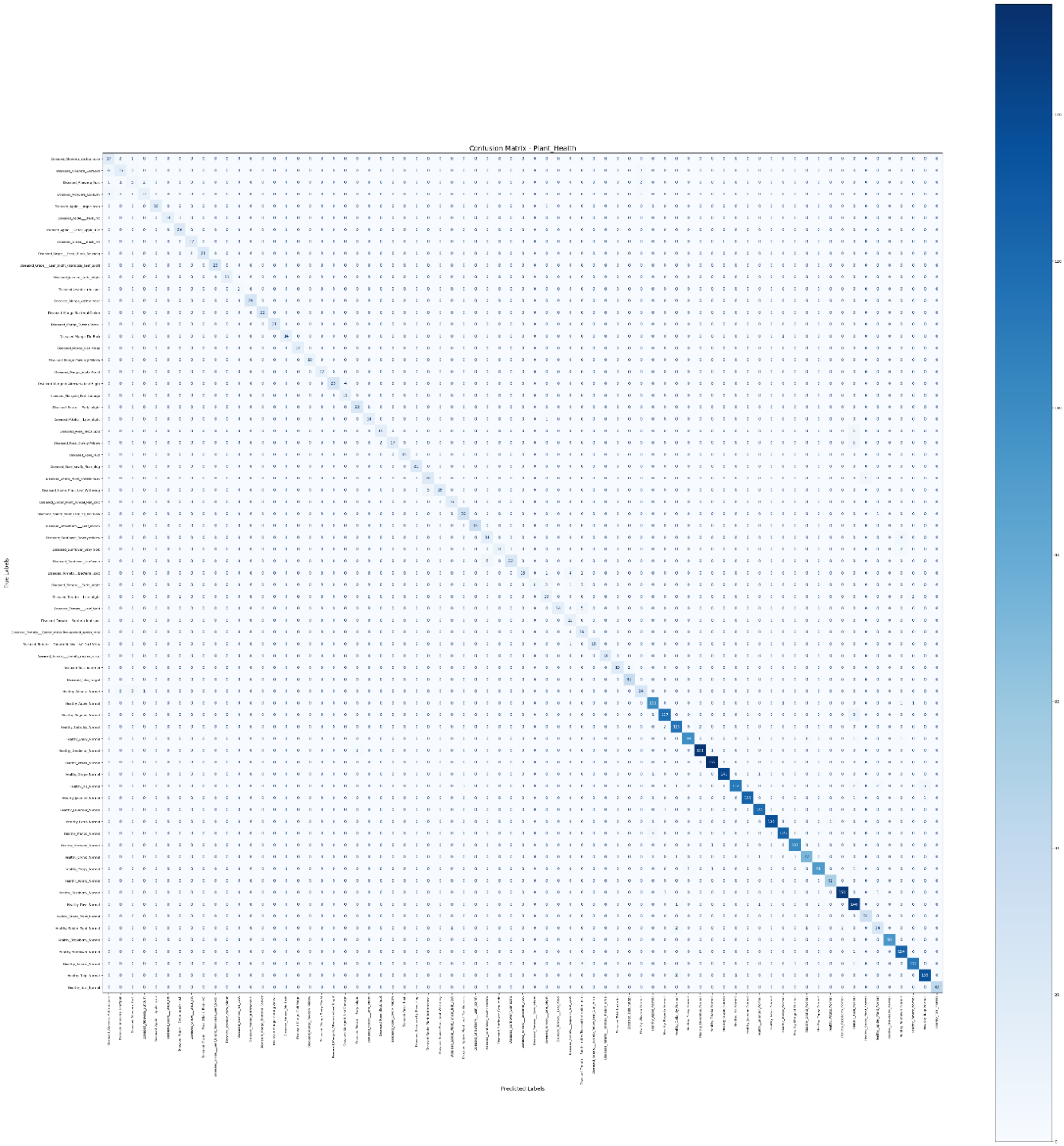
This figure presents the Confusion Matrix for the Plant Species model, evaluating classification accuracy across 26 scientific plant species names. The clear, dark blue diagonal elements indicate a highly accurate model, with most samples correctly classified under their respective true labels. The highest concentrations of correct predictions occur in classes like *Mangifera indica* (249) and *Solanum lycopersicum* (240). Off-diagonal misclassifications are minimal and isolated, showing only minor confusion between closely related botanical targets (e.g., *Phalaenopsis spp* occasionally confused with *Rosa spp*).

Fig 4.5.1.3: Confusion Matrix for Plant Organ



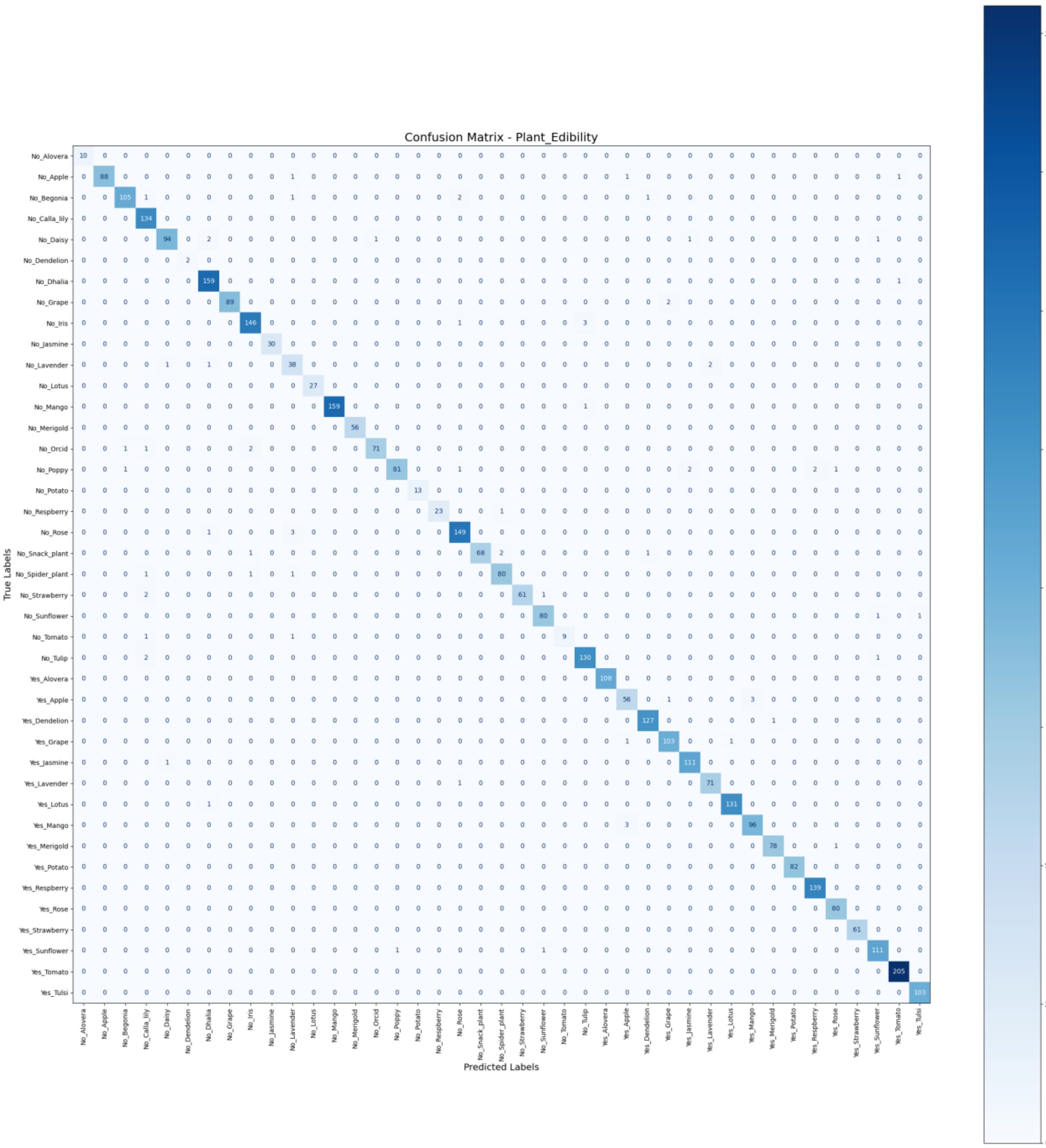
This figure displays the Confusion Matrix for the Plant Organ model, evaluating classification performance across a highly complex dataset of 98 distinct categories. Despite the significantly larger number of target classes, the matrix exhibits a distinct, continuous diagonal line of blue squares stretching from the top-left to the bottom-right. This pattern indicates that the model maintains exceptional precision and high true-positive rates even as task complexity scales up. The surrounding off-diagonal space remains heavily populated by zeros and light pixels, demonstrating that cross-class misclassification and noise are kept to a minimum across nearly 100 plant organ categories.

Fig 4.5.1.4: Confusion Matrix for Plant Health



This figure displays the Confusion Matrix for the Plant Health model, which evaluates performance across 71 distinct health and disease conditions. The matrix demonstrates strong predictive accuracy, characterized by the clearly defined diagonal line of blue squares tracking true positive outcomes from top-left to bottom-right. While the model maintains robust overall performance, the off-diagonal areas reveal slightly more visible noise and minor scattered misclassifications compared to previous models. This expected variation corresponds directly with the model's overall accuracy metric of **94.97%**, reflecting the inherent difficulty of distinguishing between subtle, visually overlapping symptoms of plant diseases across a large number of classes.

Fig 4.5.1.5: Confusion Matrix for Plant Edibility



This figure displays the Confusion Matrix for the Plant Edibility model, evaluating classification performance across 41 target categories split into non-edible ("No") and edible ("Yes") variants of different plants. The matrix demonstrates outstanding classification accuracy, marked by a sharp, unbroken diagonal line of blue squares tracking true positive predictions from the top-left to the bottom-right corner. The model handles the distinct binary state across multiple plant types effortlessly, with the highest concentration of correct predictions visible in the *Yes_Tomato* category (205). Off-diagonal spaces are nearly clean and populated almost entirely by zeros, indicating that cross-class confusion between edible and non-edible classes is virtually non-existent.

4.5.2 Classification Report Analysis

The classification report provides an evaluation of the model's performance using precision, recall, and F1-score across all classes. The results demonstrate that the ConvNeXtBase model produces no errors when identifying images from different types of plants and therefore is highly accurate in identifying types of plants. F1-scores also show that classifications are performed at a high level across all plant categories as well as that there are only a few instances of class misclassification due to the similarities in the appearance of certain classes. Overall, the ConvNeXtBase model has provided accurate and reliable classification capability.

Table 4.5.2.1: Classification Report for Plant Name

Plant Name	Precision	Recall	F1-Score	Support
Alovera	1.00	0.99	1.00	105
Apple	0.97	0.97	0.97	147
Begonia	0.99	0.99	0.99	119
Calla_lily	0.95	0.97	0.96	118
Daisy	0.95	0.98	0.97	104
Dendelion	0.98	0.99	0.99	150
Dhalia	0.99	0.98	0.99	176
Grape	1.00	0.99	0.99	211
Iris	0.96	0.98	0.97	135
Jasmine	0.97	0.95	0.96	132
Lavender	0.98	0.94	0.96	127
Lotus	0.99	0.99	0.99	149
Mango	1.00	1.00	1.00	237
Merigold	1.00	0.99	0.99	160
Orcid	0.99	0.94	0.96	77

Plant Name	Precision	Recall	F1-Score	Support
Poppy	0.99	0.95	0.97	79
Potato	0.98	0.99	0.98	95
Respberry	0.99	1.00	1.00	152
Rose	0.96	0.99	0.97	233
Snack_plant	1.00	0.99	0.99	74
Spider_plant	0.99	0.97	0.98	78
Strawberry	0.98	1.00	0.99	128
Sunflower	1.00	0.98	0.99	190
Tomato	0.99	1.00	0.99	231
Tulip	0.98	0.98	0.98	128
Tulsi	1.00	0.99	1.00	103
Accuracy			**0.98**	**3638**
Macro Avg	**0.98**	**0.98**	**0.98**	**3638**
Weighted Avg	**0.98**	**0.98**	**0.98**	**3638**

This table presents the detailed Classification Report for the Plant Name model, evaluating its predictive quality across 26 distinct plant categories using total test support of 3,638 samples. The model demonstrates outstanding and balanced performance, achieving an overall Accuracy, Macro Average, and Weighted Average of 98% across Precision, Recall, and F1-Score. Individual class metrics highlight exceptional precision and reliability, with categories such as Alovera, Mango, Tulip, and Tulsi hitting perfect or near-perfect scores of 1.00 in multiple core metrics. Even the lowest-performing categories, such as Calla lily and Lavender, maintain highly robust F1-Scores of 96%, underscoring the model's strong capability to classify diverse plant types accurately with minimal errors.

Table 4.5.2.2: Classification Report for Plant Species

Plant Species	Precision	Recall	F1-Score	Support
Aloe barbadensis miller	1.00	1.00	1.00	105
Begonia spp	0.99	0.95	0.97	108
Bellis perennis	0.98	0.95	0.97	107
Chlorophytum comosum	0.99	0.99	0.99	77
Dahlia pinnata	0.97	1.00	0.98	150

Plant Species	Precision	Recall	F1-Score	Support
Dracaena trifasciata	1.00	0.96	0.98	90
Fragaria × ananassa	1.00	0.98	0.99	128
Helianthus annuus	0.99	1.00	0.99	208
Iris germanica	0.97	0.99	0.98	144
Jasminum sambac	0.98	0.98	0.98	127
Lavandula angustifolia	0.97	0.99	0.98	116
Malus domestica	0.99	0.96	0.98	160
Mangifera indica	1.00	0.99	0.99	251
Nelumbo nucifera	0.98	1.00	0.99	137
Ocimum tenuiflorum	1.00	1.00	1.00	88
Papaver rhoeas	0.96	0.98	0.97	102
Phalaenopsis spp	0.98	0.86	0.92	73
Rosa spp	0.98	0.99	0.98	215
Rubus idaeus	0.99	1.00	1.00	151
Solanum lycopersicum	0.99	0.99	0.99	243
Solanum tuberosum	0.96	0.98	0.97	99
Tagetes erecta	0.99	0.99	0.99	156
Taraxacum officinale	0.99	0.99	0.99	141
Tulipa spp	0.96	0.99	0.97	139
Vitis vinifera	0.99	0.99	0.99	187
Zantedeschia aethiopica	0.96	1.00	0.98	136
Accuracy			**0.98**	**3638**
Macro Avg	**0.98**	**0.98**	**0.98**	**3638**
Weighted Avg	**0.98**	**0.98**	**0.98**	**3638**

This table outlines the Classification Report for the Plant Species model, detailing its evaluation across 26 distinct scientific botanical categories using a test dataset of 3,638 samples. The model demonstrates top-tier classification reliability, achieving an overall Accuracy, Macro Average, and Weighted Average of 98% across Precision, Recall, and F1-Score. Highly precise performance is evident in classes like *Aloe barbadensis miller*, *Ocimum tenuiflorum*, and *Rubus idaeus*, which achieve perfect or near-perfect scores of 1.00 across multiple metrics. The lowest outlier in the dataset is *Phalaenopsis spp*, which exhibits a slightly lower recall of 86% and an

F1-Score of 92%, reflecting minor sensitivity issues with that specific category, though the model remains exceptionally strong overall.

Table 4.5.2.3: Classification Report for Plant Organ

Organ Type	**Precision**	**Recall**	**F1-Score**	**Support**
flower				
flower_Alovera	1.00	1.00	1.00	1
flower_Apple	1.00	1.00	1.00	9
flower_Begonia	1.00	0.99	0.99	86
flower_Calla_lily	0.98	1.00	0.99	80
flower_Dahlia	1.00	1.00	1.00	122
flower_Daisy	0.99	0.97	0.98	98
flower_Dendelion	1.00	1.00	1.00	96
flower_Grape	1.00	1.00	1.00	9
flower_Iris	1.00	1.00	1.00	103
flower_Jasmine	0.99	0.98	0.99	103
flower_Lavender	1.00	1.00	1.00	83
flower_Lotus	1.00	1.00	1.00	115
flower_Mango	1.00	1.00	1.00	10
flower_Marigold	0.97	1.00	0.99	78
flower_Orchid	1.00	0.99	0.99	71
flower_Poppy	0.97	0.96	0.97	76
flower_Potato	1.00	1.00	1.00	13
flower_Respberry	1.00	0.40	0.57	5
flower_Rose	0.98	1.00	0.99	88
flower_Snack_plant	1.00	1.00	1.00	4
flower_Spider_plant	0.40	1.00	0.57	2
flower_Strawberry	1.00	1.00	1.00	9
flower_Sunflower	1.00	0.99	0.99	99
flower_Tomato	1.00	1.00	1.00	6
flower_Tulip	0.98	1.00	0.99	106
flower_Tulsi	0.00	0.00	0.00	0

Organ Type	Precision	Recall	F1-Score	Support
fruit				
fruit_Apple	0.98	0.97	0.97	59
fruit_Grape	0.97	1.00	0.99	110
fruit_Mango	0.97	0.99	0.98	95
fruit_Potato	0.97	1.00	0.98	30
fruit_Respberry	1.00	1.00	1.00	111
fruit_Strawberry	1.00	0.98	0.99	59
fruit_Tomato	1.00	1.00	1.00	101
leaf				
leaf_Alovera	1.00	1.00	1.00	98
leaf_Apple	1.00	1.00	1.00	64
leaf_Begonia	1.00	0.96	0.98	24
leaf_Calla_lily	1.00	1.00	1.00	14
leaf_Dendelion	0.96	0.92	0.94	25
leaf_Dhalia	1.00	1.00	1.00	17
leaf_Grape	1.00	1.00	1.00	79
leaf_Iris	0.90	0.82	0.86	11
leaf_Jasmine	1.00	1.00	1.00	29
leaf_Lavender	1.00	0.95	0.98	22
leaf_Lotus	1.00	1.00	1.00	29
leaf_Mango	0.99	1.00	1.00	128
leaf_Merigold	1.00	1.00	1.00	66
leaf_Orcid	0.83	1.00	0.91	5
leaf_Potato	0.99	0.97	0.98	72
leaf_Respberry	1.00	1.00	1.00	16
leaf_Rose	1.00	1.00	1.00	102
leaf_Snack_plant	1.00	1.00	1.00	72
leaf_Spider_plant	1.00	1.00	1.00	78
leaf_Strawberry	1.00	0.98	0.99	57
leaf_Sunflower	1.00	1.00	1.00	68
leaf_Tomato	0.98	1.00	0.99	146

Organ Type	Precision	Recall	F1-Score	Support
leaf_Tulip	0.85	0.92	0.88	12
leaf_Tulsi	1.00	1.00	1.00	71
root				
root_Alovera	1.00	1.00	1.00	8
root_Apple	1.00	0.67	0.80	3
root_Begonia	0.75	1.00	0.86	3
root_Calla_lily	0.80	0.89	0.84	9
root_Dendelion	0.92	1.00	0.96	11
root_Dhalia	0.85	0.92	0.88	12
root_Grape	0.00	0.00	0.00	3
root_Iris	0.91	0.91	0.91	11
root_Jasmine	0.00	0.00	0.00	1
root_Lavender	0.75	1.00	0.86	6
root_Lotus	1.00	1.00	1.00	7
root_Merigold	0.00	0.00	0.00	1
root_Orchid	1.00	0.50	0.67	2
root_Poppy	0.00	0.00	0.00	1
root_Potato	1.00	1.00	1.00	3
root_Rose	0.78	1.00	0.88	21
root_Snack_plant	1.00	0.67	0.80	3
root_Spider_plant	1.00	0.33	0.50	3
root_Strawberry	1.00	1.00	1.00	4
root_Sunflower	1.00	0.82	0.90	11
root_Tomato	1.00	1.00	1.00	4
root_Tulip	1.00	1.00	1.00	10
root_Tulsi	1.00	1.00	1.00	9
seed				
seed_Apple	1.00	1.00	1.00	3
seed_Begonia	0.67	0.67	0.67	3
seed_Calla_lily	0.90	1.00	0.95	9
seed_Daisy	0.00	0.00	0.00	2

Organ Type	Precision	Recall	F1-Score	Support
seed_Dendelion	1.00	1.00	1.00	3
seed_Dhalia	0.73	1.00	0.85	11
seed_Grape	1.00	1.00	1.00	7
seed_Iris	1.00	1.00	1.00	11
seed_Lavender	1.00	0.86	0.92	14
seed_Lotus	0.95	1.00	0.97	19
seed_Mango	1.00	0.60	0.75	5
seed_Merigold	1.00	1.00	1.00	3
seed_Orcid	1.00	1.00	1.00	7
seed_Poppy	1.00	0.67	0.80	6
seed_Rose	1.00	0.92	0.96	12
seed_Sunflower	1.00	1.00	1.00	16
seed_Tomato	1.00	0.75	0.86	4
seed_Tulip	1.00	1.00	1.00	15
Accuracy			**0.98**	**3638**
Macro Avg	**0.90**	**0.89**	**0.89**	**3638**
Weighted Avg	**0.98**	**0.98**	**0.98**	**3638**

This table presents the detailed Classification Report for the Plant Organ model, evaluating predictive metrics across various organ types (flower, fruit, leaf, root, and seed) sub-classified by plant name. The model delivers an outstanding overall Accuracy and Weighted Average of 98%, demonstrating highly precise classification across the majority of the 3,638 test samples. The leaf and fruit organ categories exhibit near-perfect performance, with dominant classes like *leaf_Tomato* (146 samples) and *fruit_Respberry* (111 samples) consistently scoring at or close to 1.00 across Precision, Recall, and F1-Score.

However, the Macro Average drops to 89% due to performance variations in classes with extremely low sample support. For instance, classes with minimal data points—such as *flower_Spider_plant* (support of 2), *root_Grape* (support of 3), or *seed_Daisy* (support of 2)—show significantly compromised scores due to data scarcity. Despite these isolated dataset imbalances impacting the macro average, the model demonstrates robust capability in extracting complex, fine-grained organ features under sufficient training data

Table 4.5.2.4: Classification Report for Plant Health

Health	Precision	Recall	F1-Score	Support
Diseased_Aloevera_Anthracnose	0.61	0.85	0.71	20
Diseased_Aloevera_LeafSpot	0.68	0.60	0.64	25
Diseased_Aloevera_Rust	0.42	0.45	0.43	11
Diseased_Aloevera_Sunburn	0.79	0.58	0.67	19
Diseased_Apple___Apple_scab	1.00	0.89	0.94	18
Diseased_Apple___Black_rot	1.00	0.93	0.96	14
Diseased_Apple___Cedar_apple_rust	0.91	1.00	0.95	20
Diseased_Grape___Black_rot	0.89	0.94	0.92	18
Diseased_Grape___Esca_(Black_Measles)	1.00	0.91	0.95	23
Diseased_Grape___Leaf_blight_(Isariopsis_Leaf_Spot)	1.00	1.00	1.00	23
Diseased_Jesmine_early_blight	1.00	1.00	1.00	11
Diseased_Jesmine_red_spot	1.00	1.00	1.00	2
Diseased_Mango_Anthracnose	1.00	1.00	1.00	26
Diseased_Mango_Bacterial Canker	1.00	1.00	1.00	22
Diseased_Mango_Cutting Weevil	1.00	1.00	1.00	21
Diseased_Mango_Die Back	1.00	0.93	0.97	15
Diseased_Mango_Gall Midge	1.00	1.00	1.00	14
Diseased_Mango_Powdery Mildew	1.00	1.00	1.00	10
Diseased_Mango_Sooty Mould	1.00	1.00	1.00	12
Diseased_Marigold_Alternaria Leaf Blight	1.00	0.86	0.93	29
Diseased_Marigold_Pest Damage	0.73	1.00	0.85	11
Diseased_Potato___Early_blight	0.92	0.96	0.94	23
Diseased_Potato___Late_blight	0.88	0.93	0.90	15
Diseased_Rose_Black Spot	0.88	0.75	0.81	20
Diseased_Rose_Downy Mildew	1.00	0.71	0.83	24
Diseased_Rose_Rust	1.00	1.00	1.00	15
Diseased_Rose_sawfly_Rose_slug	1.00	1.00	1.00	21
Diseased_Snake_Plant_Anthracnose	0.80	0.80	0.80	25

Health	Precision	Recall	F1-Score	Support
Diseased_Snake_Plant_Leaf_Withering	1.00	0.78	0.88	23
Diseased_Spider_Plant_Fungal_leaf_spot	0.89	1.00	0.94	16
Diseased_Spider_Plant_Leaf_Tip_Necrosis	0.97	0.94	0.95	32
Diseased_Strawberry___Leaf_scorch	1.00	1.00	1.00	33
Diseased_Sunflower___Downy mildew	0.74	0.74	0.74	19
Diseased_Sunflower_Gray mold	1.00	0.77	0.87	13
Diseased_Sunflower_Leaf scars	0.92	0.81	0.86	27
Diseased_Tomato___Bacterial_spot	1.00	0.71	0.83	21
Diseased_Tomato___Early_blight	1.00	0.46	0.63	13
Diseased_Tomato___Late_blight	0.62	0.71	0.67	14
Diseased_Tomato___Leaf_Mold	1.00	0.74	0.85	19
Diseased_Tomato___Septoria_leaf_spot	0.65	1.00	0.79	11
Diseased_Tomato___Spider_mites Two-spotted_spider_mite	0.57	1.00	0.73	16
Diseased_Tomato___Tomato_Yellow_Leaf_Curl_Virus	1.00	0.94	0.97	16
Diseased_Tomato___Tomato_mosaic_virus	1.00	1.00	1.00	18
Diseased_Tulsi_bacterial	1.00	0.90	0.95	21
Diseased_Tulsi_fungal	0.95	1.00	0.97	37
Healthy_Alovera_Normel	0.83	0.80	0.81	30
Healthy_Apple_Normal	0.92	0.97	0.94	104
Healthy_Begonia_Normal	0.97	0.91	0.94	128
Healthy_Calla_lily_Normal	0.94	0.97	0.95	119
Healthy_Daisy_Normal	0.98	0.96	0.97	103
Healthy_Dendelion_Normal	0.98	0.98	0.98	154
Healthy_Dhalia_Normal	0.98	1.00	0.99	155
Healthy_Grape_Normal	0.99	0.99	0.99	143
Healthy_Iris_Normal	0.99	0.91	0.95	123
Healthy_Jasmine_Normal	0.98	0.98	0.98	128
Healthy_Lavender_Normal	0.97	0.99	0.98	121

Health	Precision	Recall	F1-Score	Support
Healthy_Lotus_Normal	0.99	0.98	0.99	139
Healthy_Mango_Normal	0.98	0.97	0.97	129
Healthy_Merigold_Normal	0.98	0.99	0.99	103
Healthy_Orchid_Normal	0.97	0.94	0.95	77
Healthy_Poppy_Normal	0.99	0.96	0.97	95
Healthy_Potato_Normal	0.98	1.00	0.99	52
Healthy_Raspberry_Normal	0.99	0.99	0.99	152
Healthy_Rose_Normal	0.85	0.98	0.91	151
Healthy_Snake_Plant_Normal	0.86	1.00	0.93	31
Healthy_Spider_Plant_Normal	0.86	0.83	0.84	29
Healthy_Strawberry_Normal	1.00	1.00	1.00	93
Healthy_Sunflower_Normel	0.94	0.98	0.96	126
Healthy_Tomato_Normal	0.97	0.96	0.97	117
Healthy_Tulip_Normal	0.96	0.98	0.97	138
Healthy_Tulsi_Normal	1.00	1.00	1.00	42
accuracy			**0.95**	**3638**
macro avg	**0.92**	**0.91**	**0.91**	**3638**
weighted avg	**0.95**	**0.95**	**0.95**	**3638**

This table presents the detailed Classification Report for the Plant Health model, which evaluates predictive metrics across 71 distinct categories covering both healthy and specific diseased states across various plant types, using 3,638 total test samples. The model demonstrates strong overall performance, securing an Accuracy and Weighted Average of 95%, alongside a Macro Average F1-Score of 91%. Healthy plant categories consistently yield exceptional results, with classes like *Healthy_Strawberry_Normal* and *Healthy_Tulsi_Normal* achieving perfect 1.00 scores across Precision, Recall, and F1-Score due to their distinct visual characteristics. Numerous specific disease conditions also achieve perfect scores, such as *Diseased_Mango_Anthracnose* and *Diseased_Grape_Leaf_blight*. Conversely, performance variations are observed in challenging disease sub-classes. For instance, *Diseased_Aloevera_Rust* drops to a low F1-Score of 0.43 (0.42 Precision, 0.45 Recall), and *Diseased_Tomato_Early_blight* exhibits a low recall of 0.46 despite its perfect precision. These drops highlight classification challenges induced by low sample support and highly overlapping visual symptoms between closely related leaf spot and blight variations, though the model remains highly robust at scale.

Table 4.5.2.4: Classification Report for Plant Edibility

Edibility	Precision	Recall	F1-Score	Support
No_Alovera	1.00	1.00	1.00	10
No_Apple	1.00	0.97	0.98	91
No_Begonia	0.98	0.95	0.97	110
No_Calla_lily	0.94	1.00	0.97	134
No_Daisy	0.98	0.95	0.96	99
No_Dendelion	1.00	1.00	1.00	2
No_Dhalia	0.97	0.99	0.98	160
No_Grape	1.00	0.98	0.99	91
No_Iris	0.97	0.97	0.97	150
No_Jasmine	1.00	1.00	1.00	30
No_Lavender	0.84	0.90	0.87	42
No_Lotus	1.00	1.00	1.00	27
No_Mango	1.00	0.99	1.00	160
No_Merigold	1.00	1.00	1.00	56
No_Orcid	0.99	0.95	0.97	75
No_Poppy	0.99	0.92	0.95	88
No_Potato	1.00	1.00	1.00	13
No_Respberry	1.00	0.96	0.98	24
No_Rose	0.97	0.97	0.97	153
No_Snack_plant	1.00	0.94	0.97	72
No_Spider_plant	0.96	0.96	0.96	83
No_Strawberry	1.00	0.95	0.98	64
No_Sunflower	0.98	0.98	0.98	82
No_Tomato	1.00	0.82	0.90	11
No_Tulip	0.97	0.98	0.97	133
Yes_Alovera	1.00	1.00	1.00	108

Health	**Precision**	**Recall**	**F1-Score**	**Support**
Yes_Apple	0.92	0.93	0.93	60
Yes_Dendelion	0.98	0.99	0.99	128
Yes_Grape	0.97	0.98	0.98	105
Yes_Jasmine	0.97	0.99	0.98	112
Yes_Lavender	0.97	0.99	0.98	72
Yes_Lotus	0.99	0.99	0.99	132
Yes_Mango	0.97	0.97	0.97	99
Yes_Merigold	0.99	0.99	0.99	79
Yes_Potato	1.00	1.00	1.00	82
Yes_Respberry	0.99	1.00	0.99	139
Yes_Rose	0.98	1.00	0.99	80
Yes_Strawberry	1.00	1.00	1.00	61
Yes_Sunflower	0.97	0.98	0.98	113
Yes_Tomato	0.99	1.00	1.00	205
Yes_Tulsi	0.99	1.00	1.00	103
Accuracy			**0.98**	**3638**
Macro Avg	**0.98**	**0.97**	**0.98**	**3638**
Weighted Avg	**0.98**	**0.98**	**0.98**	**3638**

This table details the Classification Report for the Plant Edibility model, evaluating its predictive success across 41 binary categories split into non-edible ("No") and edible ("Yes") states of various plants, using a total test support of 3,638 samples. The model demonstrates excellent reliability, achieving a high overall Accuracy, Macro Average, and Weighted Average of 98% across the major evaluation criteria. Numerous individual classes yield perfect 1.00 performance, notably *Yes_Tomato* (with the highest support of 205 samples), *Yes_Strawberry*, and *No_Potato*, demonstrating the framework's capability to differentiate plant parts based on edibility rules.

Minor performance drops are concentrated in a few non-edible categories, such as *No_Lavender*, which exhibits the lowest metrics with an F1-Score of 87% (0.84 Precision, 0.90 Recall), and *No_Tomato*, which shows a recall of 82%. These slight variances point to occasional visual overlaps between specific toxic and non-toxic leaf features, though the system maintains high precision and consistency overall.

4.6 Discussion

The deep learning models developed for this research were able to achieve a high level of accuracy when identifying plants from images. Specifically, ConvNeXtBase has consistently shown to be more accurate than ResNet50, EfficientNetB3, and EfficientNetV2L with respect to accuracy, feature extraction, and generalization. This demonstrates that ConvNeXtBase is a robust option for multi-task classification of plants. ConvNeXtBase's high level of performance was attributable to access to a curated, diverse data set and optimized training methods.

4.7 Conclusion

The results of the study in this Chapter present the experimental data and analysis of different DL-based models that are intended for classifying plants. ConvNeXtBase was identified as the best model based on the evaluation metrics of accuracy/loss curves, ROC curves, confusion matrices, and classification reports all of which demonstrate very good ability to predict multiple attributes (plant name, species, organs, edibility and health) accurately. The findings from this research support the idea that DL can be used for automated identification of plants, making a significant contribution towards the development of precision agriculture and environmental monitoring industries.

Chapter 5
Conclusion and Future Work

5.1 Introduction

This Chapter presents a synthesis of the research contributions produced by the DL-based system for automated plant characterization through images, focusing on species, organs, edibility, and overall health of plants. Additionally, an evaluation of architectural benchmarks is performed on ConvNeXtBase against other significantly similar architectures (ResNet50, EfficientNetB3 and EfficientNetV2L). Results from the experimentation confirm that these models can successfully identify salient visual characteristics leading to high rates of correct classifications. Discussion in the chapter also includes potential impacts upon society and its environment, ethical implications, limitations of the study, and suggestions for future improvements to automated plant identification.

5.2 Impact on Society, Environment, and Sustainability

The DL model of plant identification systems has hastened the accurate identification of plant species and diseases, increasing agricultural crop yield, minimizing losses and assisting in providing food security to farmers, researchers and agronomists. The availability of access to plant identification information democratizes access to this knowledge, including students, botanists and the public. The accurate identification of plant species and health allows for better assessment of biodiversity and conservation efforts targeted at endangered species, which can provide information regarding the appropriate interventions to apply in specific ecosystems; the timely identification of plant diseases leads to a reduced need for the application of chemical pesticides, thus creating more sustainable and ecologically responsible agricultural practices.

5.3 Ethical Issues

Deep learning technologies can deliver plenty of benefits. However, there are several ethical considerations which should be considered when developing these types of systems. The first ethical consideration is data privacy and ownership. Data collected from various sources, including photographic images taken of plants, need to be treated with respect, and if copyrighted images are being used, they must be done so with permission from the original creator. The second ethical consideration is the reliability and accuracy of the model. An automated system making poor predictions due to incorrect models can lead to inaccurate decision-making by farmers when trying to identify food plant species or whether a crop has a disease. For this reason, any such system should be treated as an adjunct support system to supplement the expertise of a trained person rather than as a complete replacement for this expertise.

5.4 Current Limitations & Future Work

While there were multiple successes regarding the initial deep learning model of plant identification via ConvNeXtBase, there were also limitations that indicate further potential for continued research to improve the current model and serve additional purposes in the future.

Limitations:

Although a promising deep learning classification system exists for this research, several limitations remain. The model's accuracy and ability to generalize to unrecorded plants are restricted by a limited number of image samples and plant species in the training dataset. Furthermore, overlapping visual characteristics among certain species make precise differentiation difficult. Environmental factors during image capture such as lighting, noise, and quality can also degrade prediction accuracy. Finally, the system suffers from limited functionality, as it relies solely on images and fails to incorporate crucial contextual data like geographical location or seasonal indicators.

Future Work:

To address these limitations and enhance the system's utility, future studies will focus on several key areas. First, expanding the dataset by increasing the number of plant species, image varieties, and growth stages will help the model learn complex features and improve reliability. Second, investigating advanced deep learning architectures and hybrid techniques will optimize feature extraction and general classification. Third, efforts will be directed at enhancing species-level identification to better differentiate closely related plants with similar visual traits. Finally, the system's value will be elevated by including less common plant types for biodiversity tracking and providing practical information on the medicinal, nutritional, and agricultural uses of each species.

5.5 Conclusion

This research study has demonstrated the successful implementation of deep learning, resulting in ConvNeXtBase, for multi-task classification of plants from images. Information collected include species classification, organ classification, edible classification, and health status of the plant. The experiments undertaken show that this method works as a better method for extracting features and predicting classifications than other techniques currently being used. The results of this study will help to improve precision farming, provide a means of monitoring global biodiversity, and aid in studies of digital botany, which were all specific goals outlined in this study. If expanded and improved datasets are developed, this methodology can continue to provide useful information to farmers, researchers and conservationists in the future.

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